

###Dependencies

```
# Install Ultralytics (for YOLOv8)
```

```
!pip install ultralytics roboflow
```

```
Collecting ultralytics
```

```
  Downloading ultralytics-8.3.96-py3-none-any.whl.metadata (35 kB)
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Collecting roboflow
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  Downloading roboflow-1.1.58-py3-none-any.whl.metadata (9.7 kB)
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Requirement already satisfied: numpy<=2.1.1,>=1.23.0 in  
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Requirement already satisfied: matplotlib>=3.3.0 in  
/usr/local/lib/python3.11/dist-packages (from ultralytics) (3.10.0)
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Requirement already satisfied: opencv-python>=4.6.0 in  
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Requirement already satisfied: pillow>=7.1.2 in  
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Requirement already satisfied: pyyaml>=5.3.1 in  
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Requirement already satisfied: requests>=2.23.0 in  
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Requirement already satisfied: scipy>=1.4.1 in  
/usr/local/lib/python3.11/dist-packages (from ultralytics) (1.14.1)
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Requirement already satisfied: torch>=1.8.0 in  
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(2.6.0+cu124)
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(0.21.0+cu124)
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Requirement already satisfied: kiwisolver>=1.3.1 in
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Attempting uninstall: nvidia-cublas-cu12
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Attempting uninstall: idna
Found existing installation: idna 3.10

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Uninstalling idna-3.10:
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Attempting uninstall: nvidia-cudnn-cu12
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Attempting uninstall: nvidia-cusolver-cu12
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12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nvidia-cuda-nvrtc-cu12-
12.4.127 nvidia-cuda-runtime-cu12-12.4.127 nvidia-cudnn-cu12-9.1.0.70
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nvjitlink-cu12-12.4.127 opencv-python-headless-4.10.0.84 pillow-heif-
0.22.0 python-dotenv-1.1.0 roboflow-1.1.58 ultralytics-8.3.96
ultralytics-thop-2.0.14

```

###Custom Training PyTorch YOLOv8 Model

Retrieving Training Dataset

```

# Import necessary libraries
from roboflow import Roboflow

```

```

rf = Roboflow(api_key="")
project = rf.workspace("sample-").project("ui-")
version = project.version(2)
dataset = version.download("yolov8")

```

Collecting ultralytics

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  Downloading ultralytics-8.3.95-py3-none-any.whl.metadata (35 kB)

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Collecting roboflow

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  Downloading roboflow-1.1.58-py3-none-any.whl.metadata (9.7 kB)

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Requirement already satisfied: numpy<=2.1.1,>=1.23.0 in
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Requirement already satisfied: matplotlib>=3.3.0 in
/usr/local/lib/python3.11/dist-packages (from ultralytics) (3.10.0)
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(2.6.0+cu124)
Requirement already satisfied: torchvision>=0.9.0 in
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Collecting nvidia-cufft-cu12==11.2.1.3 (from torch>=1.8.0-  
>ultralytics)  
Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-  
manylinux2014_x86_64.whl.metadata (1.5 kB)  
Collecting nvidia-curand-cu12==10.3.5.147 (from torch>=1.8.0-  
>ultralytics)  
Downloading nvidia_curand_cu12-10.3.5.147-py3-none-  
manylinux2014_x86_64.whl.metadata (1.5 kB)  
Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch>=1.8.0-  
>ultralytics)  
Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-  
manylinux2014_x86_64.whl.metadata (1.6 kB)  
Collecting nvidia-cusparselt-cu12==0.6.2 (from torch>=1.8.0-  
>ultralytics)  
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in  
/usr/local/lib/python3.11/dist-packages (from torch>=1.8.0-  
>ultralytics) (0.6.2)  
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in  
/usr/local/lib/python3.11/dist-packages (from torch>=1.8.0-  
>ultralytics) (2.21.5)  
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in  
/usr/local/lib/python3.11/dist-packages (from torch>=1.8.0-  
>ultralytics) (12.4.127)  
Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch>=1.8.0-  
>ultralytics)  
Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-  
manylinux2014_x86_64.whl.metadata (1.5 kB)  
Requirement already satisfied: triton==3.2.0 in  
/usr/local/lib/python3.11/dist-packages (from torch>=1.8.0-  
>ultralytics) (3.2.0)  
Requirement already satisfied: sympy==1.13.1 in  
/usr/local/lib/python3.11/dist-packages (from torch>=1.8.0-  
>ultralytics) (1.13.1)  
Requirement already satisfied: mpmath<1.4,>=1.1.0 in  
/usr/local/lib/python3.11/dist-packages (from sympy==1.13.1-  
>torch>=1.8.0->ultralytics) (1.3.0)  
Requirement already satisfied: MarkupSafe>=2.0 in  
/usr/local/lib/python3.11/dist-packages (from jinja2->torch>=1.8.0-  
>ultralytics) (3.0.2)  
Downloading ultralytics-8.3.95-py3-none-any.whl (949 kB)  
949.8/949.8 kB 19.5 MB/s eta
```

```

0:00:00
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0:00:00
----- 66.8/66.8 kB 5.9 MB/s eta
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anylinux_2_17_x86_64.manylinux2014_x86_64.whl (49.9 MB)
----- 49.9/49.9 MB 14.8 MB/s eta
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anylinux_2_17_x86_64.manylinux2014_x86_64.whl (7.8 MB)
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anylinux2014_x86_64.whl (21.1 MB)
----- 21.1/21.1 MB 55.6 MB/s eta
0:00:00
e-cul2, nvidia-cuda-nvrtc-cul2, nvidia-cuda-cupti-cul2, nvidia-cublas-
cul2, idna, nvidia-cuspars-cul2, nvidia-cudnn-cul2, nvidia-cusolver-
cul2, roboflow, ultralytics-thop, ultralytics
  Attempting uninstall: opencv-python-headless
    Found existing installation: opencv-python-headless 4.11.0.86
    Uninstalling opencv-python-headless-4.11.0.86:
      Successfully uninstalled opencv-python-headless-4.11.0.86
  Attempting uninstall: nvidia-nvjitlink-cul2

```

```
Found existing installation: nvidia-nvjitlink-cu12 12.5.82
Uninstalling nvidia-nvjitlink-cu12-12.5.82:
  Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82
Attempting uninstall: nvidia-curand-cu12
Found existing installation: nvidia-curand-cu12 10.3.6.82
Uninstalling nvidia-curand-cu12-10.3.6.82:
  Successfully uninstalled nvidia-curand-cu12-10.3.6.82
Attempting uninstall: nvidia-cufft-cu12
Found existing installation: nvidia-cufft-cu12 11.2.3.61
Uninstalling nvidia-cufft-cu12-11.2.3.61:
  Successfully uninstalled nvidia-cufft-cu12-11.2.3.61
Attempting uninstall: nvidia-cuda-runtime-cu12
Found existing installation: nvidia-cuda-runtime-cu12 12.5.82
Uninstalling nvidia-cuda-runtime-cu12-12.5.82:
  Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82
Attempting uninstall: nvidia-cuda-nvrtc-cu12
Found existing installation: nvidia-cuda-nvrtc-cu12 12.5.82
Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:
  Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82
Attempting uninstall: nvidia-cuda-cupti-cu12
Found existing installation: nvidia-cuda-cupti-cu12 12.5.82
Uninstalling nvidia-cuda-cupti-cu12-12.5.82:
  Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82
Attempting uninstall: nvidia-cublas-cu12
Found existing installation: nvidia-cublas-cu12 12.5.3.2
Uninstalling nvidia-cublas-cu12-12.5.3.2:
  Successfully uninstalled nvidia-cublas-cu12-12.5.3.2
Attempting uninstall: idna
Found existing installation: idna 3.10
Uninstalling idna-3.10:
  Successfully uninstalled idna-3.10
Attempting uninstall: nvidia-cuspars-cu12
Found existing installation: nvidia-cuspars-cu12 12.5.1.3
Uninstalling nvidia-cuspars-cu12-12.5.1.3:
  Successfully uninstalled nvidia-cuspars-cu12-12.5.1.3
Attempting uninstall: nvidia-cudnn-cu12
Found existing installation: nvidia-cudnn-cu12 9.3.0.75
Uninstalling nvidia-cudnn-cu12-9.3.0.75:
  Successfully uninstalled nvidia-cudnn-cu12-9.3.0.75
Attempting uninstall: nvidia-cusolver-cu12
Found existing installation: nvidia-cusolver-cu12 11.6.3.83
Uninstalling nvidia-cusolver-cu12-11.6.3.83:
  Successfully uninstalled nvidia-cusolver-cu12-11.6.3.83
Successfully installed filetype-1.2.0 idna-3.7 nvidia-cublas-cu12-
12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nvidia-cuda-nvrtc-cu12-
12.4.127 nvidia-cuda-runtime-cu12-12.4.127 nvidia-cudnn-cu12-9.1.0.70
nvidia-cufft-cu12-11.2.1.3 nvidia-curand-cu12-10.3.5.147 nvidia-
cusolver-cu12-11.6.1.9 nvidia-cuspars-cu12-12.3.1.170 nvidia-
nvjitlink-cu12-12.4.127 opencv-python-headless-4.10.0.84 pillow-heif-
0.22.0 python-dotenv-1.0.1 roboflow-1.1.58 ultralytics-8.3.95
```

```
ultralytics-thop-2.0.14
loading Roboflow workspace...
loading Roboflow project...
```

```
Downloading Dataset Version Zip in UI-2 to yolov8:: 100%|██████████|
577/577 [00:00<00:00, 2563.37it/s]
```

```
Extracting Dataset Version Zip to UI-2 in yolov8:: 100%|██████████|
120/120 [00:00<00:00, 9395.32it/s]
```

```
Creating new Ultralytics Settings v0.0.6 file []
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
https://docs.ultralytics.com/quickstart/#ultralytics-settings.
Downloading
https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8s
.pt to 'yolov8s.pt'...
100% 21.5M/21.5M [00:00<00:00, 197MB/s]
Ultralytics 8.3.95 [] Python-3.11.11 torch-2.6.0+cu124 CUDA:0 (Tesla
T4, 15095MiB)
engine/trainer: task=detect, mode=train, model=yolov8s.pt,
data=/content/UI-2/data.yaml, epochs=100, time=None, patience=100,
batch=16, imgsz=640, save=True, save_period=-1, cache=False, device=0,
workers=8, project=None, name=train, exist_ok=False, pretrained=True,
optimizer=auto, verbose=True, seed=0, deterministic=True,
single_cls=False, rect=False, cos_lr=False, close_mosaic=10,
resume=False, amp=True, fraction=1.0, profile=False, freeze=None,
multi_scale=False, overlap_mask=True, mask_ratio=4, dropout=0.0,
val=True, split=val, save_json=False, save_hybrid=False, conf=None,
iou=0.7, max_det=300, half=False, dnn=False, plots=True, source=None,
vid_stride=1, stream_buffer=False, visualize=False, augment=False,
agnostic_nms=False, classes=None, retina_masks=False, embed=None,
show=False, save_frames=False, save_txt=False, save_conf=False,
save_crop=False, show_labels=True, show_conf=True, show_boxes=True,
line_width=None, format=torchscript, keras=False, optimize=False,
int8=False, dynamic=False, simplify=True, opset=None, workspace=None,
nms=False, lr0=0.01, lrf=0.01, momentum=0.937, weight_decay=0.0005,
warmup_epochs=3.0, warmup_momentum=0.8, warmup_bias_lr=0.1, box=7.5,
cls=0.5, dfl=1.5, pose=12.0, kobj=1.0, nbs=64, hsv_h=0.015, hsv_s=0.7,
hsv_v=0.4, degrees=0.0, translate=0.1, scale=0.5, shear=0.0,
perspective=0.0, flipud=0.0, fliplr=0.5, bgr=0.0, mosaic=1.0,
mixup=0.0, copy_paste=0.0, copy_paste_mode=flip,
auto_augment=randaugment, erasing=0.4, crop_fraction=1.0, cfg=None,
tracker=botsort.yaml, save_dir=runs/detect/train
Downloading https://ultralytics.com/assets/Arial.ttf to
```

```

'/root/.config/Ultralytics/Arial.ttf'...
100% 755k/755k [00:00<00:00, 20.0MB/s]
WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
E0000 00:00:1742783906.894429    1333 cuda_dnn.cc:8310] Unable to
register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
E0000 00:00:1742783906.960607    1333 cuda_blas.cc:1418] Unable to
register cuBLAS factory: Attempting to register factory for plugin
cuBLAS when one has already been registered
Overriding model.yaml nc=80 with nc=6

```

	from	n	params	module
arguments				
0	-1	1	928	ultralytics.nn.modules.conv.Conv
[3, 32, 3, 2]				
1	-1	1	18560	ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]				
2	-1	1	29056	ultralytics.nn.modules.block.C2f
[64, 64, 1, True]				
3	-1	1	73984	ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]				
4	-1	2	197632	ultralytics.nn.modules.block.C2f
[128, 128, 2, True]				
5	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
6	-1	2	788480	ultralytics.nn.modules.block.C2f
[256, 256, 2, True]				
7	-1	1	1180672	ultralytics.nn.modules.conv.Conv
[256, 512, 3, 2]				
8	-1	1	1838080	ultralytics.nn.modules.block.C2f
[512, 512, 1, True]				
9	-1	1	656896	
ultralytics.nn.modules.block.SPPF				[512, 512, 5]
10	-1	1	0	
torch.nn.modules.upsampling.Upsample				[None, 2, 'nearest']
11	[-1, 6]	1	0	
ultralytics.nn.modules.conv.Concat				[1]
12	-1	1	591360	ultralytics.nn.modules.block.C2f
[768, 256, 1]				
13	-1	1	0	
torch.nn.modules.upsampling.Upsample				[None, 2, 'nearest']
14	[-1, 4]	1	0	
ultralytics.nn.modules.conv.Concat				[1]
15	-1	1	148224	ultralytics.nn.modules.block.C2f

```

[384, 128, 1]
16          -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
17          [-1, 12]  1      0
ultralytics.nn.modules.conv.Concat          [1]

18          -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
19          -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
20          [-1, 9]  1      0
ultralytics.nn.modules.conv.Concat          [1]

21          -1  1   1969152  ultralytics.nn.modules.block.C2f
[768, 512, 1]
22          [15, 18, 21]  1   2118370
ultralytics.nn.modules.head.Detect          [6, [128, 256, 512]]

```

Model summary: 129 layers, 11,137,922 parameters, 11,137,906 gradients, 28.7 GFLOPs

Transferred 349/355 items from pretrained weights

TensorBoard: Start with 'tensorboard --logdir runs/detect/train', view at <http://localhost:6006/>

Freezing layer 'model.22.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading

<https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt> to 'yolo11n.pt'...

100% 5.35M/5.35M [00:00<00:00, 80.6MB/s]

AMP: checks passed

train: Scanning /content/UI-2/train/labels... 44 images, 0 backgrounds, 0 corrupt: 100% 44/44 [00:00<00:00, 1503.56it/s]

train: New cache created: /content/UI-2/train/labels.cache
 augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

val: Scanning /content/UI-2/valid/labels... 5 images, 0 backgrounds, 0 corrupt: 100% 5/5 [00:00<00:00, 655.05it/s]

val: New cache created: /content/UI-2/valid/labels.cache

Plotting labels to runs/detect/train/labels.jpg...

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...

optimizer: AdamW(lr=0.001, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)

TensorBoard: model graph visualization added

Image sizes 640 train, 640 val

Using 2 dataloader workers

Logging results to runs/detect/train
Starting training for 100 epochs...

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	1/100	3.78G	1.851	6.907	1.31	65
640:	100% 3/3	[00:02<00:00,	1.16it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:01<00:00,	1.29s/it]		
	all	5	12	0.0323	0.125	
0.0214	0.013					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	2/100	3.78G	1.756	7.381	1.295	45
640:	100% 3/3	[00:00<00:00,	3.83it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	7.27it/s]		
	all	5	12	0.286	0.25	
0.19	0.15					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	3/100	3.79G	1.929	7.315	1.324	55
640:	100% 3/3	[00:00<00:00,	3.84it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	8.68it/s]		
	all	5	12	0.222	0.312	
0.167	0.129					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	4/100	3.84G	1.87	5.989	1.363	42
640:	100% 3/3	[00:01<00:00,	2.96it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	2.88it/s]		
	all	5	12	0.3	0.188	
0.235	0.163					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	5/100	3.87G	1.435	2.996	1.134	47
640:	100% 3/3	[00:01<00:00,	2.63it/s]			
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	4.26it/s]		
	all	5	12	0.484	0.188	
0.305	0.222					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size						

6/100	3.91G	1.25	2.296	1.035	30
640: 100%	3/3	[00:00<00:00,	3.85it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	5.53it/s]	
	all	5	12	0.362	0.329
0.395	0.255				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
7/100	3.95G	1.017	1.917	0.9708	56
640: 100%	3/3	[00:00<00:00,	3.82it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	6.70it/s]	
	all	5	12	0.89	0.5
0.576	0.406				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
8/100	3.99G	0.9843	1.757	0.9378	47
640: 100%	3/3	[00:00<00:00,	3.57it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	6.39it/s]	
	all	5	12	0.989	0.438
0.573	0.381				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
9/100	4.03G	1.06	1.611	0.9546	61
640: 100%	3/3	[00:00<00:00,	3.77it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	7.21it/s]	
	all	5	12	0.563	0.75
0.704	0.505				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
10/100	4.06G	0.9799	1.622	0.9358	53
640: 100%	3/3	[00:01<00:00,	2.84it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	3.81it/s]	
	all	5	12	0.668	0.75
0.748	0.549				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
11/100	4.1G	0.9213	1.537	0.9201	59
640: 100%	3/3	[00:01<00:00,	2.71it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00,	4.61it/s]	
	all	5	12	0.682	0.75

```

0.778      0.59
Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
12/100      4.13G      0.9953      1.583      0.944      46
640: 100% 3/3 [00:00<00:00, 3.93it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:00<00:00, 6.63it/s]
all        5        12      0.574      0.75
0.751      0.606

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
13/100      4.17G      0.9188      1.54      0.9586      44
640: 100% 3/3 [00:00<00:00, 3.58it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:00<00:00, 10.50it/s]
all        5        12      0.807      0.625
0.745      0.567

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
14/100      4.2G      1.007      1.271      0.9845      61
640: 100% 3/3 [00:00<00:00, 4.11it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:00<00:00, 6.38it/s]
all        5        12      0.616      0.871
0.777      0.597

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
15/100      4.24G      0.9186      1.099      0.9693      48
640: 100% 3/3 [00:00<00:00, 3.85it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:00<00:00, 6.54it/s]
all        5        12      0.68      0.875
0.739      0.512

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
16/100      4.31G      0.9455      1.08      0.9535      64
640: 100% 3/3 [00:00<00:00, 3.92it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:00<00:00, 7.11it/s]
all        5        12      0.745      0.869
0.85      0.649

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
17/100      4.38G      0.88      1.066      0.9197      48

```

```

640: 100% 3/3 [00:00<00:00, 3.71it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 8.63it/s]
      all          5          12      0.989      0.988
0.995      0.844

```

```

      Epoch      GPU_mem    box_loss    cls_loss    dfl_loss  Instances
Size
      18/100      4.45G      0.9654      1.159      1.001      43
640: 100% 3/3 [00:00<00:00, 3.05it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 4.64it/s]
      all          5          12      0.92      0.983
0.995      0.849

```

```

      Epoch      GPU_mem    box_loss    cls_loss    dfl_loss  Instances
Size
      19/100      4.52G      0.9396      0.9896      0.9928      61
640: 100% 3/3 [00:00<00:00, 3.43it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 8.42it/s]
      all          5          12      0.89      0.981
0.995      0.795

```

```

      Epoch      GPU_mem    box_loss    cls_loss    dfl_loss  Instances
Size
      20/100      4.59G      1.012      1.02      0.9565      57
640: 100% 3/3 [00:00<00:00, 3.86it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 8.13it/s]
      all          5          12      0.956      0.869
0.953      0.713

```

```

      Epoch      GPU_mem    box_loss    cls_loss    dfl_loss  Instances
Size
      21/100      4.62G      0.9587      1.096      0.9726      58
640: 100% 3/3 [00:00<00:00, 3.81it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 10.63it/s]
      all          5          12      0.879      0.875
0.97      0.685

```

```

      Epoch      GPU_mem    box_loss    cls_loss    dfl_loss  Instances
Size
      22/100      4.7G      0.9347      0.9575      0.9695      43
640: 100% 3/3 [00:00<00:00, 3.88it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 8.89it/s]
      all          5          12      0.883      0.873
0.97      0.761

```

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
23/100		4.73G	0.9045	1.128	0.9282	74
640:	100% 3/3	[00:00<00:00, 3.86it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 9.07it/s]		
		all	5	12	0.933	0.875
0.953	0.744					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
24/100		4.83G	0.946	1.089	0.9518	49
640:	100% 3/3	[00:00<00:00, 3.74it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 14.12it/s]		
		all	5	12	0.988	0.875
0.941	0.761					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
25/100		4.9G	0.961	0.987	0.976	56
640:	100% 3/3	[00:00<00:00, 3.83it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 9.40it/s]		
		all	5	12	0.979	0.875
0.97	0.784					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
26/100		4.97G	0.893	0.877	0.9379	59
640:	100% 3/3	[00:00<00:00, 3.99it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 6.17it/s]		
		all	5	12	0.967	1
0.995	0.797					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
27/100		5.01G	0.8934	0.8822	0.9569	50
640:	100% 3/3	[00:01<00:00, 2.94it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 8.27it/s]		
		all	5	12	0.969	0.997
0.995	0.82					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
28/100		5.08G	0.8533	0.8295	0.8908	46
640:	100% 3/3	[00:00<00:00, 3.60it/s]				

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 8.25it/s]			
	all	5	12	0.976	0.997	
0.995	0.788					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	29/100	5.15G	0.9642	0.8683	0.9715	53
640:	100% 3/3	[00:00<00:00, 3.97it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 10.62it/s]			
	all	5	12	0.976	0.997	
0.995	0.788					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	30/100	5.22G	0.903	0.9377	0.9641	38
640:	100% 3/3	[00:00<00:00, 3.85it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 9.44it/s]			
	all	5	12	0.984	1	
0.995	0.756					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	31/100	5.29G	0.9054	0.8647	0.9666	45
640:	100% 3/3	[00:00<00:00, 4.00it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 9.64it/s]			
	all	5	12	0.984	1	
0.995	0.749					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	32/100	5.39G	0.8669	0.8103	0.9644	50
640:	100% 3/3	[00:00<00:00, 3.76it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 9.05it/s]			
	all	5	12	0.907	0.999	
0.995	0.786					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	33/100	5.43G	0.9676	0.8586	0.9787	54
640:	100% 3/3	[00:00<00:00, 4.02it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 12.65it/s]			
	all	5	12	0.907	0.999	
0.995	0.786					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	34/100	5.46G	0.9009	0.8367	0.9432	41
640:	100% 3/3 [00:00<00:00, 3.96it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 10.69it/s]		
		all	5	12	0.873	0.875
0.926	0.757					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	35/100	5.53G	0.8919	0.8072	0.9902	57
640:	100% 3/3 [00:00<00:00, 3.68it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 6.53it/s]		
		all	5	12	0.93	0.868
0.933	0.721					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	36/100	5.6G	0.8882	0.829	0.9631	67
640:	100% 3/3 [00:01<00:00, 2.85it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 6.88it/s]		
		all	5	12	0.971	0.875
0.941	0.707					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	37/100	5.67G	0.8848	0.7014	0.919	66
640:	100% 3/3 [00:00<00:00, 3.93it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 8.92it/s]		
		all	5	12	0.971	0.875
0.941	0.707					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	38/100	5.74G	0.9412	0.7499	0.9838	42
640:	100% 3/3 [00:00<00:00, 3.88it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 10.94it/s]		
		all	5	12	0.965	0.99
0.995	0.772					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
	39/100	5.81G	0.8489	0.767	0.9428	64
640:	100% 3/3 [00:00<00:00, 4.00it/s]					
		Class	Images	Instances	Box(P	R

```

mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 11.45it/s]
          all          5          12          0.98          1
0.995          0.771

```

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances
Size
40/100   5.88G    0.8507   0.7519   0.9582         56
640: 100% 3/3 [00:00<00:00,  3.90it/s]
      Class    Images  Instances    Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 10.82it/s]
          all          5          12          0.99          0.963
0.995          0.74

```

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances
Size
41/100   5.92G    0.8066   0.6365   0.9119         45
640: 100% 3/3 [00:00<00:00,  4.17it/s]
      Class    Images  Instances    Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 10.38it/s]
          all          5          12          0.99          0.963
0.995          0.74

```

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances
Size
42/100   5.99G    0.8035   0.7171   0.9409         35
640: 100% 3/3 [00:00<00:00,  4.03it/s]
      Class    Images  Instances    Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00,  8.68it/s]
          all          5          12          0.991         0.892
0.988          0.665

```

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances
Size
43/100   6.05G    0.8672   0.6839   0.9519         53
640: 100% 3/3 [00:00<00:00,  4.02it/s]
      Class    Images  Instances    Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00,  9.59it/s]
          all          5          12          0.978          0.996
0.995          0.662

```

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances
Size
44/100   6.16G    0.9367   0.6641   0.9726         41
640: 100% 3/3 [00:00<00:00,  3.15it/s]
      Class    Images  Instances    Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 11.55it/s]
          all          5          12          0.981          0.983
0.995          0.724

```

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances

```

Size	45/100	6.2G	0.9036	0.6981	0.9484	45
640:	100%	3/3	[00:00<00:00, 3.05it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 7.40it/s]		
		all	5	12	0.981	0.983
0.995	0.724					

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	46/100	6.26G	0.8483	0.6031	0.9503	62
640:	100%	3/3	[00:00<00:00, 3.74it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 9.34it/s]		
		all	5	12	0.995	0.991
0.995	0.791					

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	47/100	6.33G	0.9054	0.6975	0.9705	46
640:	100%	3/3	[00:00<00:00, 3.85it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 8.64it/s]		
		all	5	12	0.989	1
0.995	0.807					

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	48/100	6.4G	0.8107	0.6704	0.9214	63
640:	100%	3/3	[00:00<00:00, 3.74it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 8.85it/s]		
		all	5	12	0.971	0.996
0.995	0.784					

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	49/100	6.44G	0.9118	0.6718	0.9567	52
640:	100%	3/3	[00:00<00:00, 4.10it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 8.84it/s]		
		all	5	12	0.971	0.996
0.995	0.784					

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	50/100	6.51G	0.8417	0.7227	0.9553	42
640:	100%	3/3	[00:00<00:00, 3.92it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 11.20it/s]		

0.995	0.771	all	5	12	0.935	1
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
51/100	6.58G	0.8183	0.6404	0.9501	61	
640: 100%	3/3	[00:00<00:00, 3.87it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95): 100%	1/1	[00:00<00:00, 9.67it/s]				
0.995	0.814	all	5	12	0.973	1
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
52/100	6.67G	0.7675	0.5815	0.9212	32	
640: 100%	3/3	[00:00<00:00, 3.79it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95): 100%	1/1	[00:00<00:00, 8.74it/s]				
0.995	0.84	all	5	12	0.978	1
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
53/100	6.71G	0.8576	0.5834	0.9086	61	
640: 100%	3/3	[00:00<00:00, 3.19it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95): 100%	1/1	[00:00<00:00, 6.30it/s]				
0.995	0.84	all	5	12	0.978	1
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
54/100	6.78G	0.8071	0.5702	0.919	45	
640: 100%	3/3	[00:01<00:00, 2.73it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95): 100%	1/1	[00:00<00:00, 5.53it/s]				
0.995	0.862	all	5	12	0.96	1
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
55/100	6.85G	0.8349	0.6131	0.9564	59	
640: 100%	3/3	[00:00<00:00, 3.89it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95): 100%	1/1	[00:00<00:00, 11.04it/s]				
0.995	0.863	all	5	12	0.949	0.942
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances

56/100	6.92G	0.8696	0.6132	0.9372	51
640: 100%	3/3	[00:00<00:00, 3.65it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 10.41it/s]		
	all	5	12	0.896	1
0.995	0.825				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	57/100	6.96G	0.8881	0.5756	48
640: 100%	3/3	[00:00<00:00, 3.96it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 10.09it/s]		
	all	5	12	0.896	1
0.995	0.825				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	58/100	7.03G	0.8166	0.5287	46
640: 100%	3/3	[00:00<00:00, 3.89it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 12.22it/s]		
	all	5	12	0.927	0.981
0.995	0.801				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	59/100	7.09G	0.8594	0.5984	44
640: 100%	3/3	[00:00<00:00, 3.82it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 8.02it/s]		
	all	5	12	0.95	0.976
0.995	0.821				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	60/100	7.16G	0.7976	0.5487	56
640: 100%	3/3	[00:00<00:00, 3.94it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 7.86it/s]		
	all	5	12	0.858	1
0.97	0.793				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	61/100	7.23G	0.8684	0.5947	74
640: 100%	3/3	[00:00<00:00, 3.92it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 11.30it/s]		
	all	5	12	0.858	1

```

0.97      0.793
Epoch    GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
62/100    7.27G      0.8094      0.5892      0.9332      65
640: 100% 3/3 [00:01<00:00, 2.87it/s]
Class     Images  Instances  Box(P      R
mAP50     mAP50-95): 100% 1/1 [00:00<00:00, 6.70it/s]
all       5        12        0.861      1
0.97      0.801

```

```

Epoch    GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
63/100    7.37G      0.8166      0.5597      0.91        60
640: 100% 3/3 [00:00<00:00, 3.01it/s]
Class     Images  Instances  Box(P      R
mAP50     mAP50-95): 100% 1/1 [00:00<00:00, 11.43it/s]
all       5        12        0.974      0.988
0.995     0.777

```

```

Epoch    GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
64/100    7.41G      0.7798      0.5528      0.925       58
640: 100% 3/3 [00:00<00:00, 3.87it/s]
Class     Images  Instances  Box(P      R
mAP50     mAP50-95): 100% 1/1 [00:00<00:00, 9.63it/s]
all       5        12        0.955      0.989
0.995     0.796

```

```

Epoch    GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
65/100    7.48G      0.847       0.5291      0.9514      62
640: 100% 3/3 [00:00<00:00, 3.78it/s]
Class     Images  Instances  Box(P      R
mAP50     mAP50-95): 100% 1/1 [00:00<00:00, 9.15it/s]
all       5        12        0.955      0.989
0.995     0.796

```

```

Epoch    GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
66/100    7.55G      0.8189      0.5464      0.9058      76
640: 100% 3/3 [00:00<00:00, 3.93it/s]
Class     Images  Instances  Box(P      R
mAP50     mAP50-95): 100% 1/1 [00:00<00:00, 11.06it/s]
all       5        12        0.947      0.991
0.995     0.782

```

```

Epoch    GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
67/100    7.58G      0.8444      0.5732      0.9449      37

```

```

640: 100% 3/3 [00:00<00:00, 3.85it/s]
      Class      Images Instances      Box(P      R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 7.60it/s]
      all          5         12      0.95      0.992
0.995      0.789

```

```

      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
Size
      68/100      7.69G      0.8145      0.5434      0.9293      47
640: 100% 3/3 [00:00<00:00, 3.81it/s]
      Class      Images Instances      Box(P      R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 14.82it/s]
      all          5         12      0.957      0.993
0.995      0.808

```

```

      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
Size
      69/100      7.76G      0.7574      0.544      0.8887      58
640: 100% 3/3 [00:00<00:00, 4.08it/s]
      Class      Images Instances      Box(P      R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 11.95it/s]
      all          5         12      0.957      0.993
0.995      0.808

```

```

      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
Size
      70/100      7.79G      0.8437      0.5738      0.943      47
640: 100% 3/3 [00:00<00:00, 4.00it/s]
      Class      Images Instances      Box(P      R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 7.97it/s]
      all          5         12      0.962      0.993
0.995      0.793

```

```

      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
Size
      71/100      7.86G      0.8175      0.5754      0.9618      46
640: 100% 3/3 [00:01<00:00, 2.92it/s]
      Class      Images Instances      Box(P      R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 4.90it/s]
      all          5         12      0.971      0.994
0.995      0.8

```

```

      Epoch      GPU_mem      box_loss      cls_loss      dfl_loss      Instances
Size
      72/100      7.93G      0.7563      0.5065      0.9127      60
640: 100% 3/3 [00:01<00:00, 2.94it/s]
      Class      Images Instances      Box(P      R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 8.93it/s]
      all          5         12      0.977      1
0.995      0.813

```

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
73/100		7.97G	0.8004	0.5176	0.9168	45
640:	100% 3/3	[00:00<00:00, 4.03it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 11.70it/s]		
		all	5	12	0.977	1
0.995	0.813					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
74/100		8.04G	0.7239	0.5119	0.905	52
640:	100% 3/3	[00:00<00:00, 3.98it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 10.25it/s]		
		all	5	12	0.985	1
0.995	0.813					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
75/100		8.11G	0.759	0.5154	0.8912	53
640:	100% 3/3	[00:00<00:00, 3.82it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 8.24it/s]		
		all	5	12	0.974	1
0.995	0.804					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
76/100		8.21G	0.7543	0.5059	0.8927	73
640:	100% 3/3	[00:00<00:00, 3.71it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 10.29it/s]		
		all	5	12	0.959	1
0.995	0.815					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
77/100		8.25G	0.7429	0.4601	0.896	51
640:	100% 3/3	[00:00<00:00, 4.06it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 10.43it/s]		
		all	5	12	0.959	1
0.995	0.815					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
78/100		8.28G	0.8195	0.494	0.9346	45
640:	100% 3/3	[00:00<00:00, 3.99it/s]				

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 12.01it/s]			
	all	5	12	0.967	1	
0.995	0.835					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	79/100	8.38G	0.7882	0.5014	0.9433	56
640:	100% 3/3	[00:00<00:00, 3.83it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 9.24it/s]			
	all	5	12	0.989	1	
0.995	0.792					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	80/100	8.46G	0.7163	0.4763	0.8955	71
640:	100% 3/3	[00:01<00:00, 2.73it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 6.93it/s]			
	all	5	12	0.988	1	
0.995	0.784					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	81/100	8.53G	0.6886	0.5029	0.8803	48
640:	100% 3/3	[00:00<00:00, 3.31it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 9.27it/s]			
	all	5	12	0.988	1	
0.995	0.784					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	82/100	8.56G	0.7454	0.4581	0.9241	48
640:	100% 3/3	[00:00<00:00, 3.87it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 12.92it/s]			
	all	5	12	0.986	1	
0.995	0.783					
	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size	83/100	8.66G	0.7133	0.474	0.9033	52
640:	100% 3/3	[00:00<00:00, 3.87it/s]				
	Class	Images	Instances	Box(P	R	
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 8.53it/s]			
	all	5	12	0.989	1	
0.995	0.775					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	84/100	8.73G	0.7723	0.4737	0.937	29
	100% 3/3 [00:00<00:00, 3.93it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 13.75it/s]		
		all	5	12	0.986	1
0.995	0.766					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	85/100	8.77G	0.7451	0.4763	0.8883	65
	100% 3/3 [00:00<00:00, 3.85it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 8.26it/s]		
		all	5	12	0.986	1
0.995	0.766					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	86/100	8.84G	0.7483	0.4838	0.9093	40
	100% 3/3 [00:00<00:00, 3.79it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 7.76it/s]		
		all	5	12	0.986	1
0.995	0.75					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	87/100	8.91G	0.768	0.4829	0.9109	38
	100% 3/3 [00:00<00:00, 3.87it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 7.84it/s]		
		all	5	12	0.991	1
0.995	0.752					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	88/100	8.98G	0.7244	0.4574	0.8886	38
	100% 3/3 [00:00<00:00, 3.30it/s]					
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 6.81it/s]		
		all	5	12	0.986	1
0.995	0.755					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	89/100	9.01G	0.7101	0.4531	0.9168	40
	100% 3/3 [00:00<00:00, 3.12it/s]					
		Class	Images	Instances	Box(P	R

```
mAP50 mAP50-95): 100% 1/1 [00:00<00:00, 4.50it/s]
all 5 12 0.986 1
0.995 0.755
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size
90/100 9.08G 0.762 0.4397 0.9148 45
640: 100% 3/3 [00:00<00:00, 3.68it/s]
Class Images Instances Box(P R
mAP50 mAP50-95): 100% 1/1 [00:00<00:00, 10.69it/s]
all 5 12 0.983 1
0.995 0.741
```

```
Closing dataloader mosaic
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, num_output_channels=3,
method='weighted_average'), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size
91/100 9.15G 0.6748 0.4142 0.8873 31
640: 100% 3/3 [00:01<00:00, 1.85it/s]
Class Images Instances Box(P R
mAP50 mAP50-95): 100% 1/1 [00:00<00:00, 3.06it/s]
all 5 12 0.982 1
0.995 0.741
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size
92/100 9.22G 0.653 0.3931 0.8814 45
640: 100% 3/3 [00:00<00:00, 4.01it/s]
Class Images Instances Box(P R
mAP50 mAP50-95): 100% 1/1 [00:00<00:00, 11.83it/s]
all 5 12 0.98 1
0.995 0.75
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size
93/100 9.29G 0.6853 0.4 0.8823 32
640: 100% 3/3 [00:00<00:00, 4.20it/s]
Class Images Instances Box(P R
mAP50 mAP50-95): 100% 1/1 [00:00<00:00, 9.33it/s]
all 5 12 0.98 1
0.995 0.75
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size
94/100 9.36G 0.6524 0.4109 0.8822 30
640: 100% 3/3 [00:00<00:00, 3.95it/s]
Class Images Instances Box(P R
```

```

mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 12.65it/s]
          all          5          12          0.979          1
0.995          0.744

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
 95/100      9.43G      0.6969      0.4359      0.8988          42
640: 100% 3/3 [00:00<00:00,  4.05it/s]
          Class      Images  Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 12.30it/s]
          all          5          12          0.979          1
0.995          0.756

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
 96/100      9.5G      0.6641      0.4035      0.891          39
640: 100% 3/3 [00:00<00:00,  3.09it/s]
          Class      Images  Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00,  8.05it/s]
          all          5          12          0.979          1
0.995          0.764

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
 97/100      9.54G      0.6515      0.3785      0.8908          35
640: 100% 3/3 [00:00<00:00,  3.10it/s]
          Class      Images  Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 14.13it/s]
          all          5          12          0.979          1
0.995          0.764

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
 98/100      9.61G      0.6443      0.4088      0.8772          34
640: 100% 3/3 [00:00<00:00,  3.93it/s]
          Class      Images  Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00,  8.97it/s]
          all          5          12          0.979          1
0.995          0.766

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
 99/100      9.68G      0.6546      0.3974      0.8569          42
640: 100% 3/3 [00:00<00:00,  3.95it/s]
          Class      Images  Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 12.51it/s]
          all          5          12          0.979          1
0.995          0.766

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances

```

Size	100/100	9.75G	0.671	0.3994	0.898	40
640:	100%	3/3	[00:00<00:00,	3.76it/s]		
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	9.31it/s]	
		all	5	12	0.979	1
0.995	0.776					

100 epochs completed in 0.045 hours.

Optimizer stripped from runs/detect/train/weights/last.pt, 22.5MB

Optimizer stripped from runs/detect/train/weights/best.pt, 22.5MB

Validating runs/detect/train/weights/best.pt...

Ultralytics 8.3.95 Python-3.11.11 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

Model summary (fused): 72 layers, 11,127,906 parameters, 0 gradients, 28.4 GFLOPs

	Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00,	17.45it/s]
	all	5	12	0.948	0.944
0.995	0.863				
	bad_logo	2	4	1	0.887
0.995	0.904				
	good_logo	5	8	0.897	1
0.995	0.822				

Speed: 0.3ms preprocess, 4.2ms inference, 0.0ms loss, 1.3ms postprocess per image

Results saved to runs/detect/train

Learn more at <https://docs.ultralytics.com/modes/train>

Training Model

Check if YOLOv8 is installed properly

```
from ultralytics import YOLO
```

Train YOLOv8 Model

```
!yolo task=detect mode=train model=yolov8s.pt data=/content/UI-2/data.yaml epochs=100 imgsz=640 batch=16 device=0
```

Retrieving Custom Trained Model and Test Dataset

Retrieving Test Dataset

```
import os
import zipfile
import gdown
import cv2
import matplotlib.pyplot as plt
from ultralytics import YOLO
```

```

# Google Drive file IDs
IMAGE_ZIP_ID = "1TBFPop4QTobKwwxCZK8gzeqfm8oplfff"
MODEL_FILE_ID = "13lJA3me2S5a9G0HYeMa8RtkCGg9XDEMj"

# Define file paths
IMAGE_ZIP_PATH = "./images.zip"
MODEL_PATH = "./yolov8_model.pt"
IMAGE_DIR = "./"

# Download images zip
gdown.download(f"https://drive.google.com/uc?id={IMAGE_ZIP_ID}",
IMAGE_ZIP_PATH, quiet=False)

# Download YOLOv8 model
gdown.download(f"https://drive.google.com/uc?id={MODEL_FILE_ID}",
MODEL_PATH, quiet=False)

# Extract images
os.makedirs(IMAGE_DIR, exist_ok=True)
with zipfile.ZipFile(IMAGE_ZIP_PATH, 'r') as zip_ref:
    zip_ref.extractall(IMAGE_DIR)

# Load YOLOv8 model
model = YOLO(MODEL_PATH)

Downloading...
From: https://drive.google.com/uc?id=1TBFPop4QTobKwwxCZK8gzeqfm8oplfff
To: /content/images.zip
100%|██████████| 341k/341k [00:00<00:00, 13.5MB/s]
Downloading...
From: https://drive.google.com/uc?id=13lJA3me2S5a9G0HYeMa8RtkCGg9XDEMj
To: /content/yolov8_model.pt
100%|██████████| 22.5M/22.5M [00:00<00:00, 132MB/s]

```

Testing Model

```

# Run inference on all images in the directory
IMAGE_DIR = "./test/"
image_files = [os.path.join(IMAGE_DIR, f) for f in
os.listdir(IMAGE_DIR) if f.endswith(('.png', '.jpg', '.jpeg'))]

for image_file in image_files:
    results = model(image_file)

# Display results
for result in results:
    img = cv2.imread(image_file)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    plt.figure(figsize=(8, 6))

```

```

        plt.imshow(result.plot()) # YOLO automatically plots
detections
        plt.axis("off")
        plt.title(f"Detections for {os.path.basename(image_file)}")
        plt.show()

```

###Detecting Clear Space Around Logo Using OpenCV and Model

```

import os
import cv2
import numpy as np
import requests
import matplotlib.pyplot as plt
from ultralytics import YOLO

# Define model paths
EAST_MODEL_PATH = "frozen_east_text_detection.pb"
EAST_MODEL_URL =
"https://raw.githubusercontent.com/oyyd/frozen\_east\_text\_detection.pb/
master/frozen\_east\_text\_detection.pb"

# Download EAST model if not present
if not os.path.exists(EAST_MODEL_PATH):
    print("Downloading EAST text detection model...")
    response = requests.get(EAST_MODEL_URL, stream=True)
    if response.status_code == 200:
        with open(EAST_MODEL_PATH, "wb") as f:
            for chunk in response.iter_content(chunk_size=1024):
                f.write(chunk)
        print("Download complete.")
    else:
        raise Exception("Failed to download the EAST model.")

# Load models
east_net = cv2.dnn.readNet(EAST_MODEL_PATH)

# Function to detect lines
def detect_and_draw_lines(image):
    """ Detects only straight horizontal and vertical lines, ensuring
    high contrast for blue headers.
    Returns detected line coordinates for further processing. """

    # Convert to HSV to isolate the blue header
    hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
    lower_blue = np.array([90, 50, 50]) # Adjust these values if
needed
    upper_blue = np.array([130, 255, 255])
    mask_blue = cv2.inRange(hsv, lower_blue, upper_blue)

    # Convert to grayscale, but prioritize the blue channel for better

```

```

contrast
    blue_channel = image[:, :, 0] # Extract blue channel
    enhanced_gray = cv2.addWeighted(blue_channel, 1.5, mask_blue, 0.5,
0) # Boost blue contrast

    # Apply adaptive thresholding for improved edge detection
    binary = cv2.adaptiveThreshold(enhanced_gray, 255,
cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                                cv2.THRESH_BINARY, 11, 2)

    # Edge detection using Canny with adaptive thresholds
    edges = cv2.Canny(binary, 30, 200) # Adjusted thresholds

    # Use Hough Line Transform to detect only straight lines
    lines = cv2.HoughLinesP(edges, rho=1, theta=np.pi/180,
threshold=80,
                                minLineLength=50, maxLineGap=5) #
Adjusted for smaller headers

    detected_lines = [] # Store detected lines

    # Draw detected lines in yellow and store them
    if lines is not None:
        for line in lines:
            x1, y1, x2, y2 = line[0]

            # Only store and draw strictly horizontal or vertical
lines
            if abs(x1 - x2) < 5 or abs(y1 - y2) < 5:
                cv2.line(image, (x1, y1), (x2, y2), (0, 255, 255), 2)
# Yellow
                detected_lines.append((x1, y1, x2, y2))

    return detected_lines # Return list of detected lines

# Function to detect text using EAST
def detect_text_east(image):
    orig = image.copy()
    (H, W) = image.shape[:2]
    newW, newH = (W // 32) * 32, (H // 32) * 32
    rW, rH = W / float(newW), H / float(newH)

    image = cv2.resize(image, (newW, newH))
    blob = cv2.dnn.blobFromImage(image, 1.0, (newW, newH), (123.68,
116.78, 103.94), swapRB=True, crop=False)
    east_net.setInput(blob)

    scores, geometry =
east_net.forward(["feature_fusion/Conv_7/Sigmoid",
"feature_fusion/concat_3"])

```

```

rects, confidences = [], []
for y in range(scores.shape[2]):
    for x in range(scores.shape[3]):
        if scores[0, 0, y, x] < 0.5:
            continue
        offsetX, offsetY = x * 4.0, y * 4.0
        angle = geometry[0, 4, y, x]
        cos, sin = np.cos(angle), np.sin(angle)
        h, w = geometry[0, 0, y, x] + geometry[0, 2, y, x],
geometry[0, 1, y, x] + geometry[0, 3, y, x]
        endX, endY = int(offsetX + (cos * geometry[0, 1, y, x]) +
(sin * geometry[0, 2, y, x])), int(offsetY - (sin * geometry[0, 1, y,
x]) + (cos * geometry[0, 2, y, x]))
        startX, startY = int(endX - w), int(endY - h)
        rects.append((startX, startY, endX, endY))
        confidences.append(float(scores[0, 0, y, x]))

boxes = cv2.dnn.NMSBoxes(rects, confidences, score_threshold=0.5,
nms_threshold=0.4)
return [(rects[i]) for i in boxes.flatten()] if len(boxes) > 0
else []

def draw_perpendicular_lines(image, ref_box, detections):
    x1, y1, x2, y2 = ref_box
    center_x, center_y = (x1 + x2) // 2, (y1 + y2) // 2

    distances = {} # Store distances for all directions

    # Define four movement directions: Up, Down, Left, Right
    directions = {
        "Up": (center_x, y1, center_x, 0), # From top boundary to
box top
        "Down": (center_x, y2, center_x, image.shape[0]), # From box
bottom to image bottom
        "Left": (x1, center_y, 0, center_y), # From left boundary to
box left
        "Right": (x2, center_y, image.shape[1], center_y) # From box
right to image right
    }

    # Convert detected lines into obstacles (small-width bounding
boxes)
    line_obstacles = []
    for (lx1, ly1, lx2, ly2) in detections:
        if abs(lx1 - lx2) < 5: # Vertical line
            line_obstacles.append((lx1 - 2, min(ly1, ly2), lx1 + 2,
max(ly1, ly2))) # Small-width box
        elif abs(ly1 - ly2) < 5: # Horizontal line
            line_obstacles.append((min(lx1, lx2), ly1 - 2, max(lx1,
lx2), ly1 + 2)) # Small-height box

```

```

# Combine text boxes, YOLO boxes, and detected lines
all_obstacles = detections + line_obstacles

for direction, (x_start, y_start, x_end, y_end) in
directions.items():
    # Initialize final end coordinates
    final_x, final_y = x_end, y_end

    # Check for obstacles in the path
    for (bx1, by1, bx2, by2) in all_obstacles:
        if "Up" in direction and bx1 <= center_x <= bx2 and by2 <
y1:
            final_y = max(final_y, by2) # Stop at the closest
component **above**
        elif "Down" in direction and bx1 <= center_x <= bx2 and
by1 > y2:
            final_y = min(final_y, by1) # Stop at the closest
component **below**
        elif "Left" in direction and by1 <= center_y <= by2 and
bx2 < x1:
            final_x = max(final_x, bx2) # Stop at the closest
component **to the right** (Fix)
        elif "Right" in direction and by1 <= center_y <= by2 and
bx1 > x2:
            final_x = min(final_x, bx1) # Stop at the closest
component **to the left**

    # Draw the final adjusted line
    cv2.line(image, (x_start, y_start), (final_x, final_y), (0,
255, 0), 2)

    # Calculate and store distance
    distance = abs(final_y - y_start) if "Up" in direction or
"Down" in direction else abs(final_x - x_start)
    distances[direction] = distance

    # Ensure text does not go out of bounds
    text_x, text_y = max(5, min(final_x, image.shape[1] - 50)),
max(15, min(final_y, image.shape[0] - 5))
    cv2.putText(image, f"{distance}px", (text_x, text_y),
cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

    bbox_size = max(x2 - x1, y2 - y1) # Take the larger dimension as
size
    min_clearance = bbox_size / 3.5 # Bounding box length of clear
space required

    print(f"Bounding Box size: {bbox_size}\tMinimum clearance:
{min_clearance}")

```

```

# Ensure bounding box size is more than 18 pixels
if bbox_size <= 18:
    print(f"Bounding Box ({x1}, {y1}, {x2}, {y2}) is too small
(Size: {bbox_size}px) - INVALID")
    return False

# Check if all distances are greater than min_clearance
if all(d > min_clearance for d in distances.values()):
    print(f"Bounding Box ({x1}, {y1}, {x2}, {y2}) - Distances:
{distances} - VALID")
    return True
else:
    print(f"Bounding Box ({x1}, {y1}, {x2}, {y2}) - Distances:
{distances} - INVALID")
    return False

# Process images
IMAGE_DIR = "./test/"
image_files = [os.path.join(IMAGE_DIR, f) for f in
os.listdir(IMAGE_DIR) if f.endswith(('.png', '.jpg', '.jpeg'))]

# Process each image
for image_file in image_files:
    img = cv2.imread(image_file)
    results = model(image_file)
    text_boxes = detect_text_fast(img)

    yolo_boxes = []
    good_logo_boxes = [] # Store multiple detections of "good_logo"

    # Detect and store straight lines
    detected_lines = detect_and_draw_lines(img) # Now stores the
lines

    for result in results[0].boxes:
        x1, y1, x2, y2 = map(int, result.xyxy[0].tolist())
        label = model.names[int(result.cls[0].item())]
        color = (255, 0, 0) if label != "good_logo" else (0, 255, 0)
        cv2.rectangle(img, (x1, y1), (x2, y2), color, 2)
        cv2.putText(img, label, (x1, y1 - 10),
cv2.FONT_HERSHEY_SIMPLEX, 0.5, color, 2)

        if label == "good_logo":
            good_logo_boxes.append((x1, y1, x2, y2)) # Append all
detections

        yolo_boxes.append((x1, y1, x2, y2))

    for (tx1, ty1, tx2, ty2) in text_boxes:

```

```

        cv2.rectangle(img, (tx1, ty1), (tx2, ty2), (0, 255, 255), 1)
# Yellow for text

    # Draw perpendicular lines for ALL detected "good_logo" instances
    for good_logo_box in good_logo_boxes:
        draw_perpendicular_lines(img, good_logo_box, yolo_boxes +
text_boxes + detected_lines)
        # Now passing detected lines too!

plt.figure(figsize=(8, 6))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.axis("off")
plt.title(f"Detections for {os.path.basename(image_file)}")
plt.show()

```

```

image 1/1 /content/test/Page 2.png: 480x640 1 bad_button, 1 good_logo,
12.9ms
Speed: 3.0ms preprocess, 12.9ms inference, 1.3ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 112      Minimum clearance: 32.0
Bounding Box (189, 40, 301, 99) - Distances: {'Up': 40, 'Down':
np.int32(33), 'Left': 189, 'Right': 1139} - VALID

```

Detections for Page 2.png



```
image 1/1 /content/test/Page 1.png: 480x640 1 good_logo, 13.0ms
Speed: 4.7ms preprocess, 13.0ms inference, 1.8ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 111      Minimum clearance: 31.714285714285715
Bounding Box (43, 37, 154, 98) - Distances: {'Up': 37, 'Down':
np.int32(33), 'Left': 43, 'Right': 1286} - VALID
```

Detections for Page 1.png



image 1/1 /content/test/Page 19.png: 480x640 4 bad_logos, 12.9ms
Speed: 2.7ms preprocess, 12.9ms inference, 1.4ms postprocess per image
at shape (1, 3, 480, 640)

Detections for Page 19.png

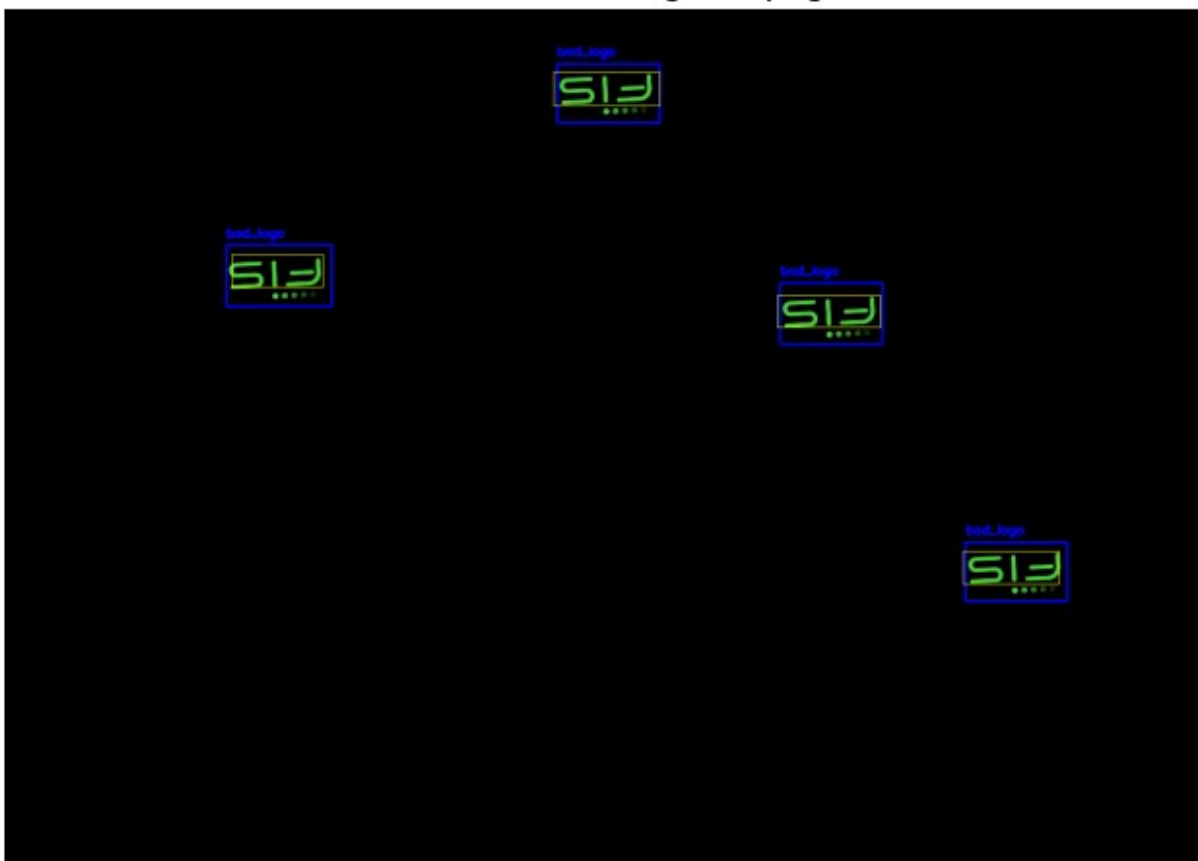


image 1/1 /content/test/Page 7.png: 480x640 1 bad_logo, 1 good_logo, 12.9ms
Speed: 3.0ms preprocess, 12.9ms inference, 1.3ms postprocess per image at shape (1, 3, 480, 640)
Bounding Box size: 120 Minimum clearance: 34.285714285714285
Bounding Box (268, 201, 388, 259) - Distances: {'Up': 201, 'Down': 765, 'Left': 268, 'Right': 1052} - VALID

Detections for Page 7.png

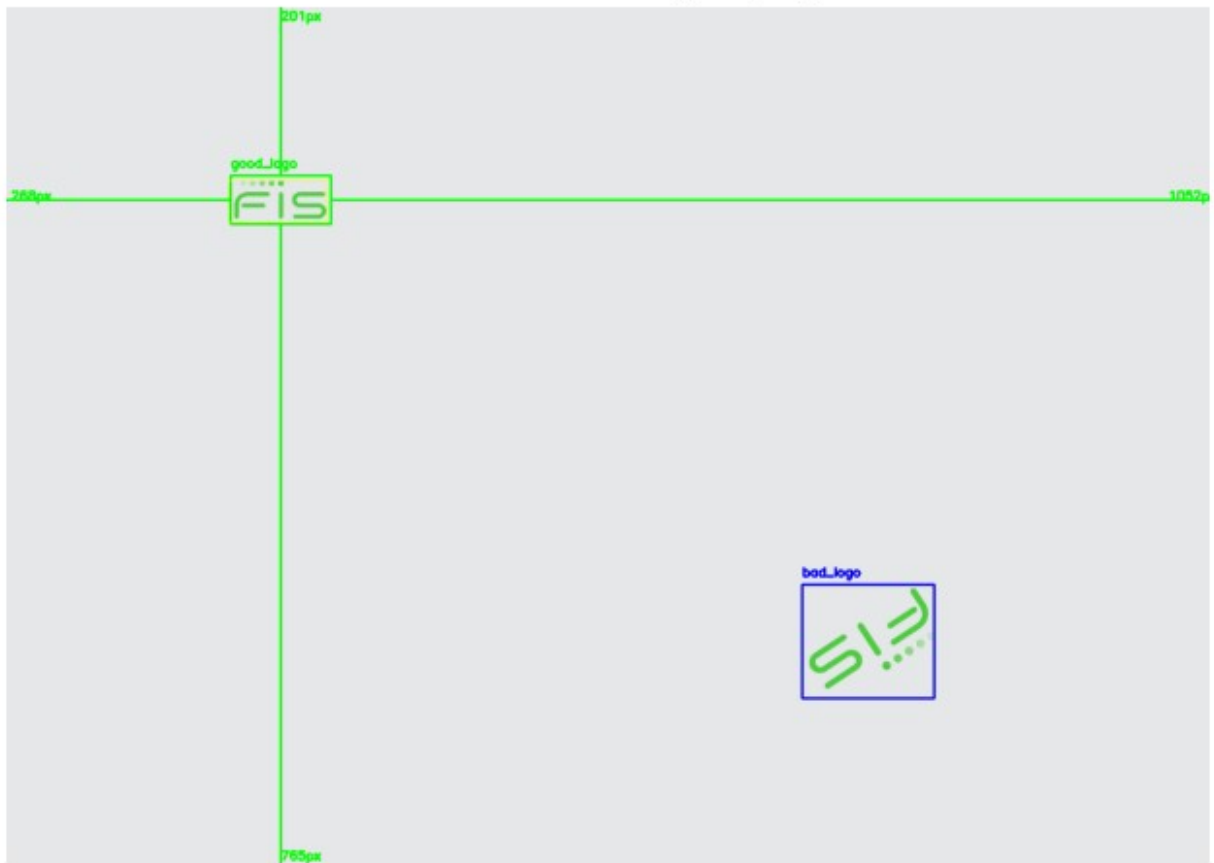
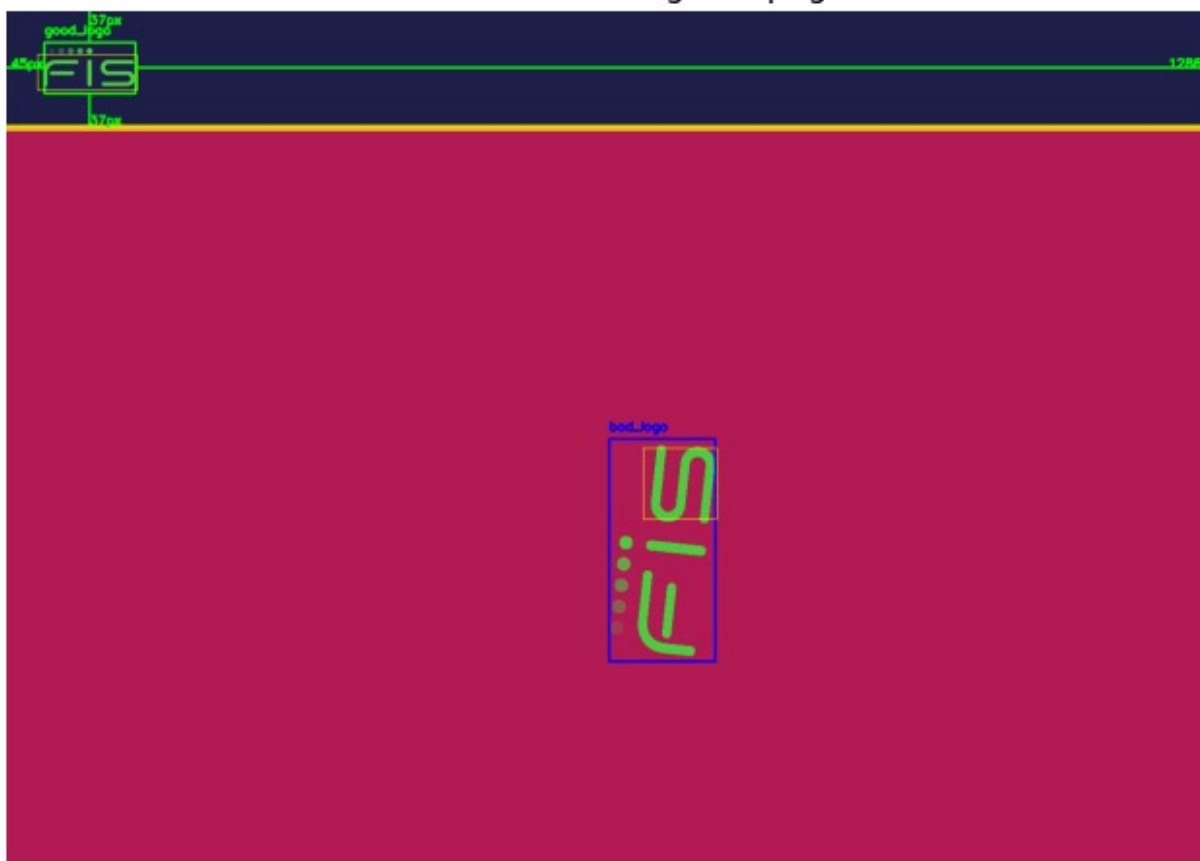


image 1/1 /content/test/Page 14.png: 480x640 1 bad_logo, 1 good_logo, 12.9ms
Speed: 2.6ms preprocess, 12.9ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)
Bounding Box size: 109 Minimum clearance: 31.142857142857142
Bounding Box (45, 37, 154, 98) - Distances: {'Up': 37, 'Down': np.int32(37), 'Left': 45, 'Right': 1286} - VALID

Detections for Page 14.png



```
image 1/1 /content/test/Page 12.png: 480x640 3 good_logos, 13.0ms
Speed: 4.7ms preprocess, 13.0ms inference, 1.4ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 117 Minimum clearance: 33.42857142857143
Bounding Box (1144, 596, 1261, 662) - Distances: {'Up': 596, 'Down':
362, 'Left': 708, 'Right': 117} - VALID
Bounding Box size: 117 Minimum clearance: 33.42857142857143
Bounding Box (316, 596, 433, 662) - Distances: {'Up': 596, 'Down':
362, 'Left': 316, 'Right': 709} - VALID
Bounding Box size: 65 Minimum clearance: 18.571428571428573
Bounding Box (1378, 596, 1440, 661) - Distances: {'Up': 596, 'Down':
363, 'Left': 114, 'Right': 0} - INVALID
```

Detections for Page 12.png

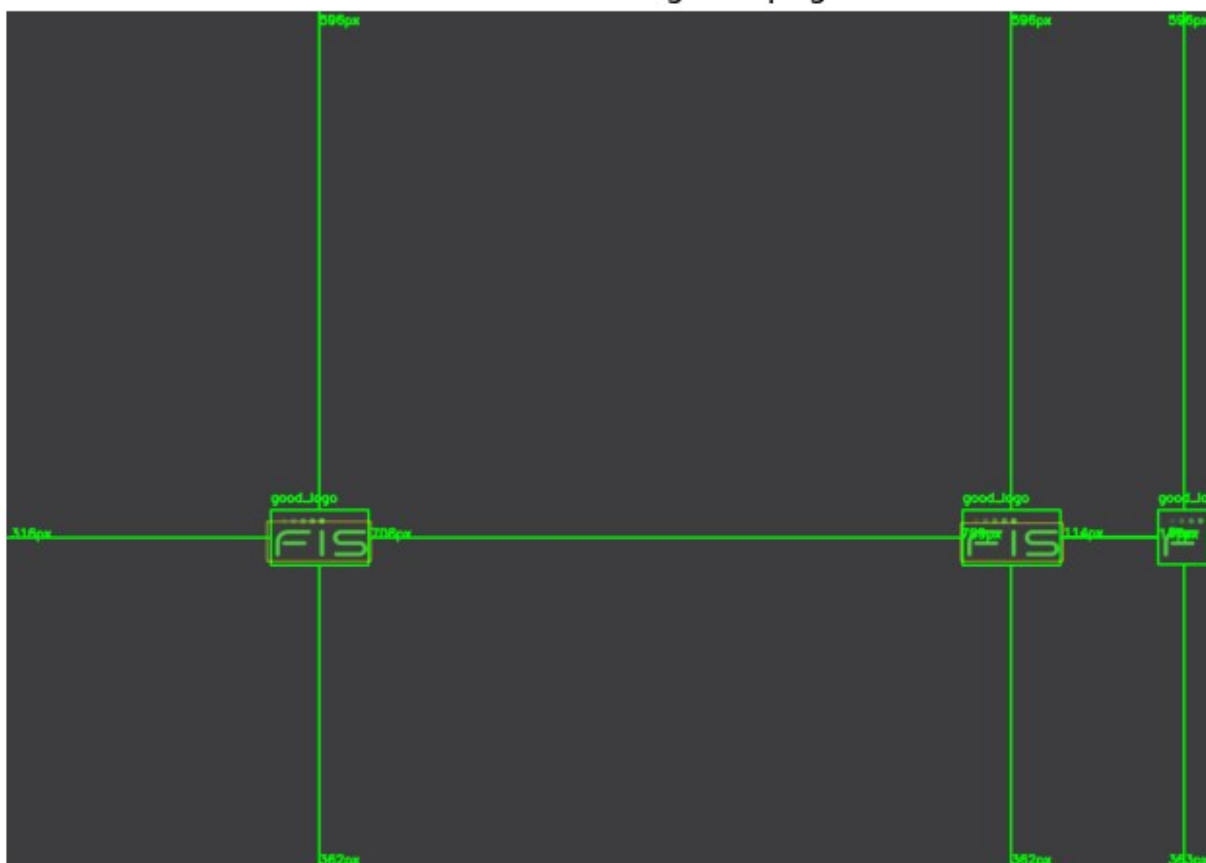
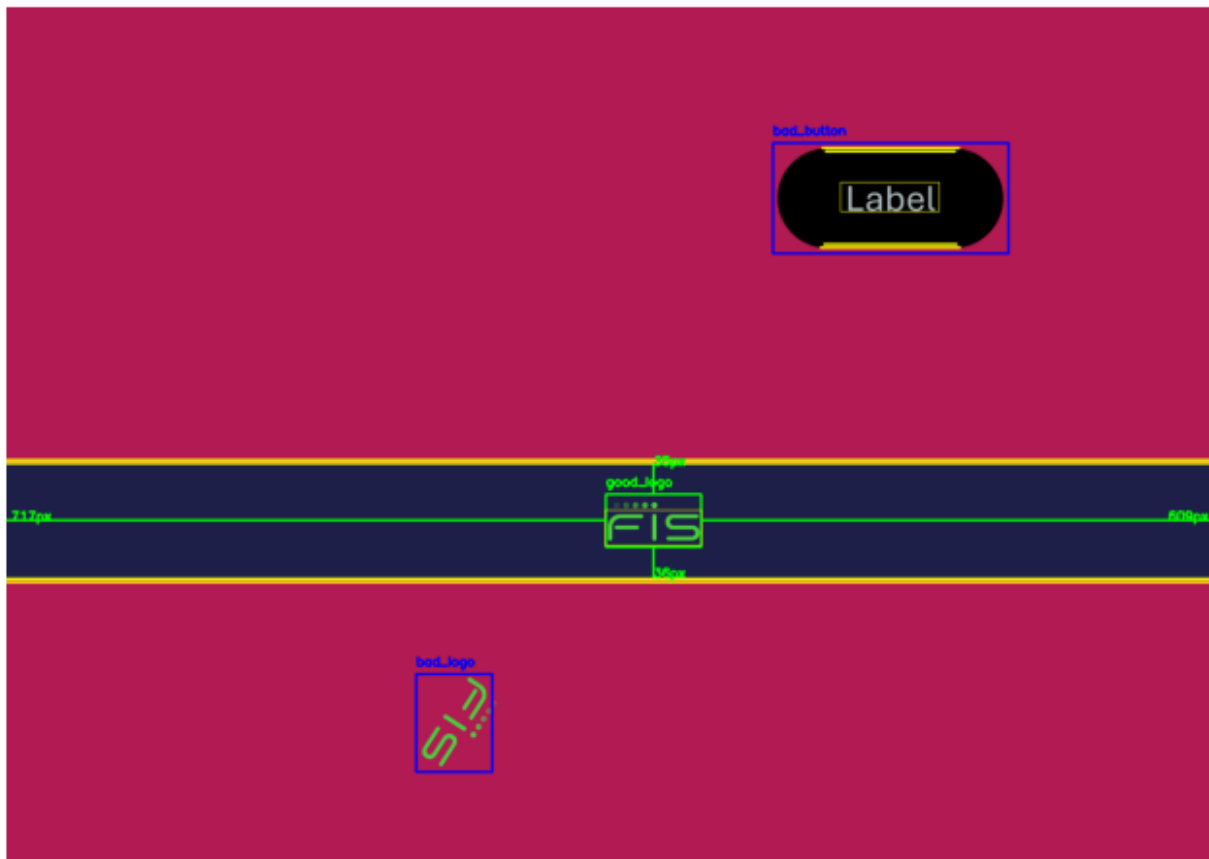


image 1/1 /content/test/Page 16.png: 480x640 1 bad_button, 1 bad_logo, 1 good_logo, 12.9ms
 Speed: 2.7ms preprocess, 12.9ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)
 Bounding Box size: 114 Minimum clearance: 32.57142857142857
 Bounding Box (717, 582, 831, 645) - Distances: {'Up': np.int32(35), 'Down': np.int32(36), 'Left': 717, 'Right': 609} - VALID

Detections for Page 16.png



```
image 1/1 /content/test/Page 11.png: 480x640 1 bad_button, 2
bad_logos, 1 good_logo, 13.0ms
Speed: 4.1ms preprocess, 13.0ms inference, 1.3ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 115      Minimum clearance: 32.857142857142854
Bounding Box (661, 455, 776, 521) - Distances: {'Up': 455, 'Down':
342, 'Left': 661, 'Right': 664} - VALID
```

Detections for Page 11.png

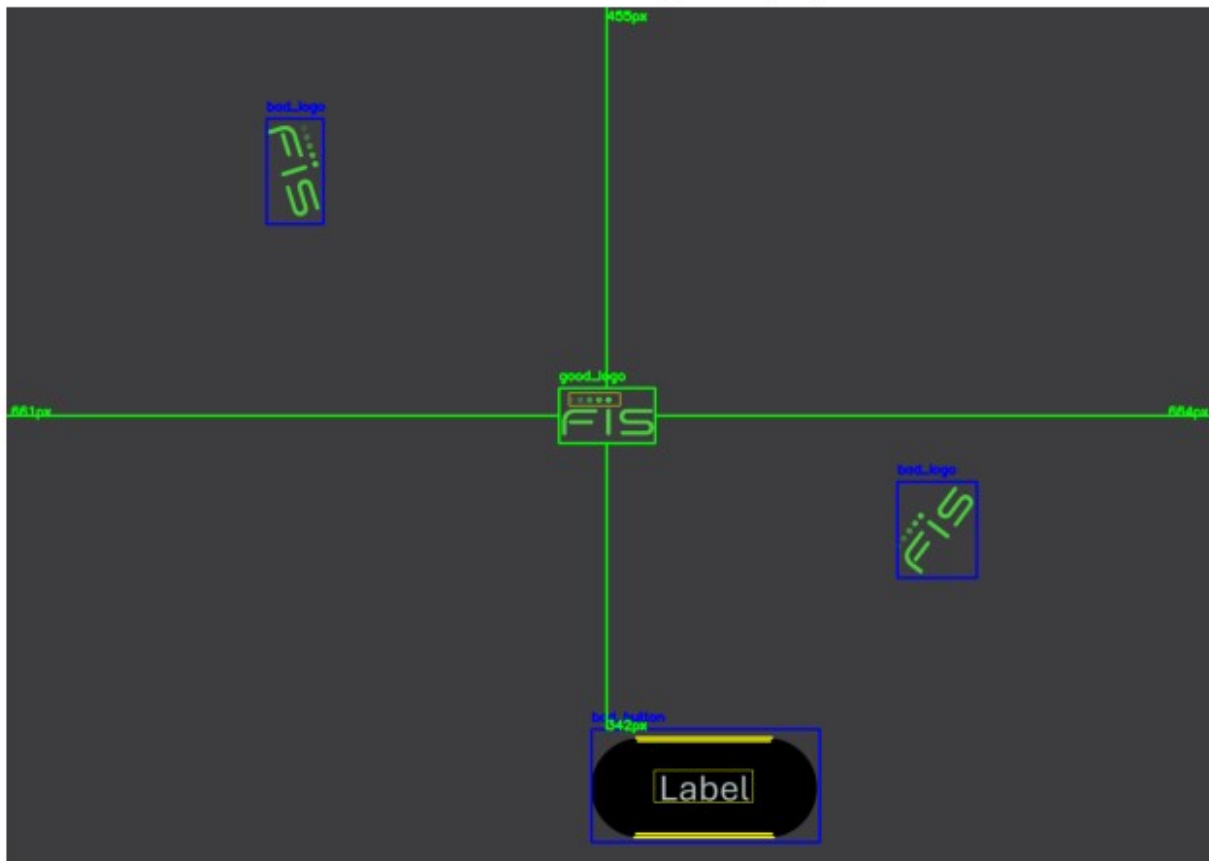
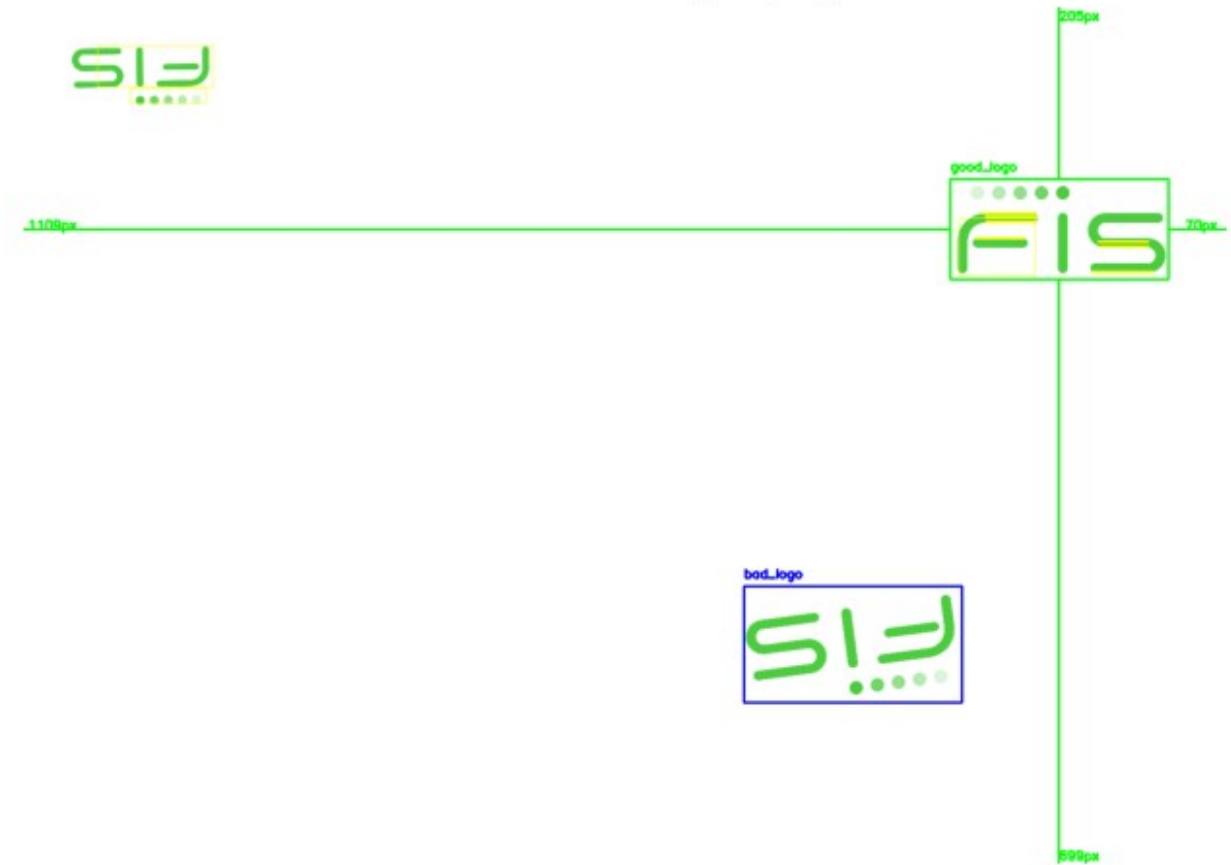


image 1/1 /content/test/Page 3.png: 480x640 1 bad_logo, 1 good_logo, 13.0ms
Speed: 2.9ms preprocess, 13.0ms inference, 1.3ms postprocess per image at shape (1, 3, 480, 640)
Bounding Box size: 261 Minimum clearance: 74.57142857142857
Bounding Box (1109, 205, 1370, 325) - Distances: {'Up': 205, 'Down': 699, 'Left': 1109, 'Right': 70} - INVALID

Detections for Page 3.png



```
image 1/1 /content/test/Page 8.png: 480x640 2 bad_logos, 3 good_logos,
13.0ms
Speed: 4.3ms preprocess, 13.0ms inference, 2.1ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 116      Minimum clearance: 33.142857142857146
Bounding Box (1203, 604, 1319, 658) - Distances: {'Up': 604, 'Down':
162, 'Left': 670, 'Right': 121} - VALID
Bounding Box size: 112      Minimum clearance: 32.0
Bounding Box (420, 580, 532, 633) - Distances: {'Up': 580, 'Down':
391, 'Left': 420, 'Right': 671} - VALID
Bounding Box size: 529      Minimum clearance: 151.14285714285714
Bounding Box (911, 837, 1440, 1024) - Distances: {'Up': 837, 'Down':
0, 'Left': 911, 'Right': 0} - INVALID
```

Detections for Page 8.png

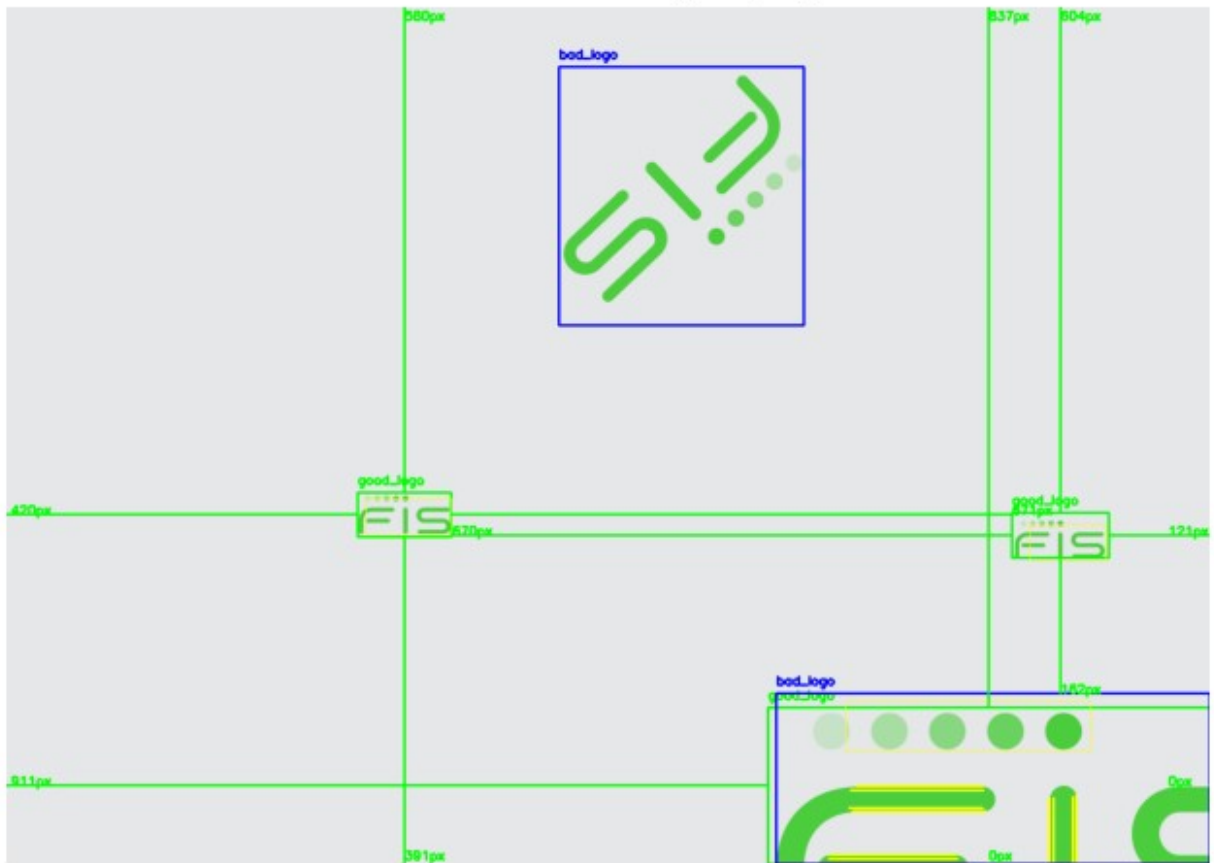
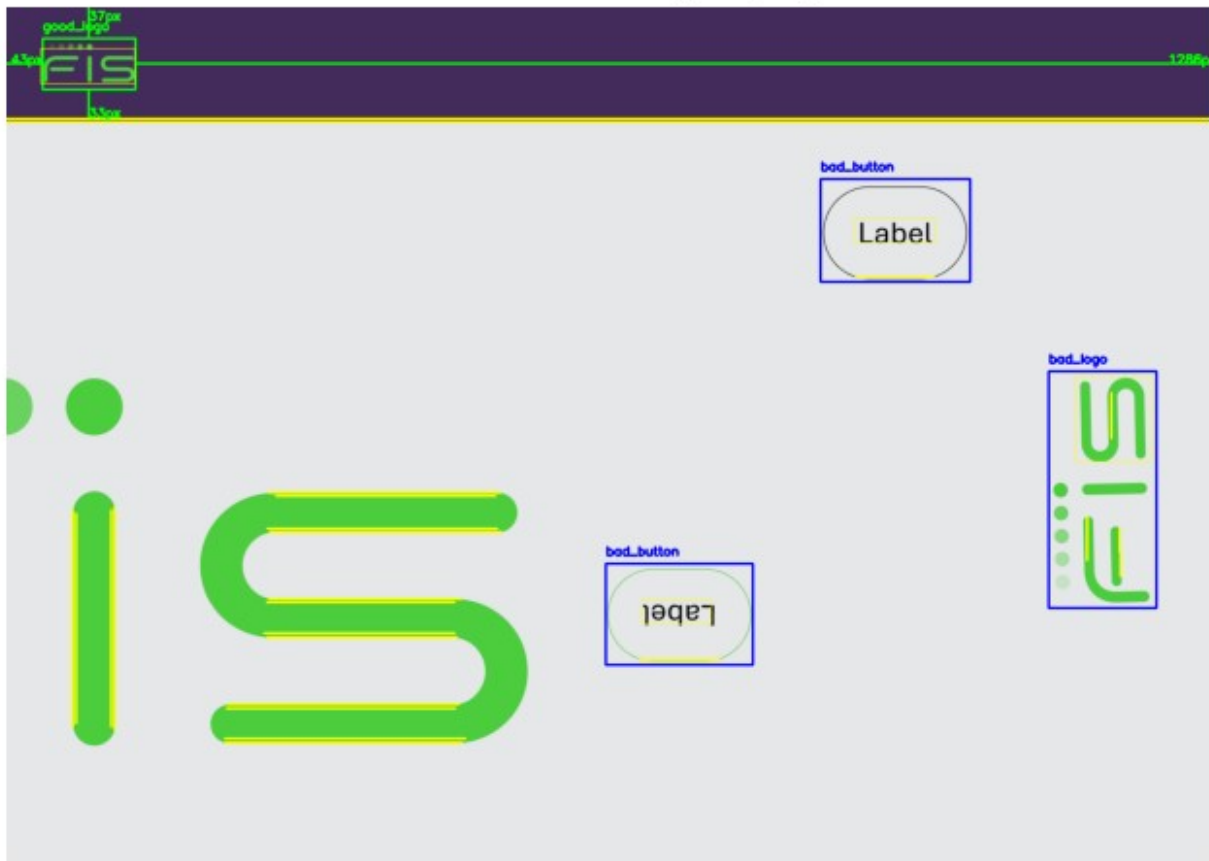


image 1/1 /content/test/Page 5.png: 480x640 2 bad_buttons, 1 bad_logo, 1 good_logo, 12.9ms
Speed: 2.8ms preprocess, 12.9ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)
Bounding Box size: 111 Minimum clearance: 31.714285714285715
Bounding Box (43, 37, 154, 98) - Distances: {'Up': 37, 'Down': np.int32(33), 'Left': 43, 'Right': 1286} - VALID

Detections for Page 5.png



```
image 1/1 /content/test/Page 17.png: 480x640 4 good_logos, 12.9ms
Speed: 2.7ms preprocess, 12.9ms inference, 1.3ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 226 Minimum clearance: 64.57142857142857
Bounding Box (457, 688, 683, 805) - Distances: {'Up': np.int32(545),
'Down': 219, 'Left': 457, 'Right': 757} - VALID
Bounding Box size: 117 Minimum clearance: 33.42857142857143
Bounding Box (1252, 900, 1369, 967) - Distances: {'Up': np.int32(757),
'Down': 57, 'Left': 1252, 'Right': 71} - VALID
Bounding Box size: 87 Minimum clearance: 24.857142857142858
Bounding Box (966, 515, 1053, 568) - Distances: {'Up': np.int32(372),
'Down': 456, 'Left': 966, 'Right': 387} - VALID
Bounding Box size: 111 Minimum clearance: 31.714285714285715
Bounding Box (43, 38, 154, 98) - Distances: {'Up': 38, 'Down':
np.int32(37), 'Left': 43, 'Right': 1286} - VALID
```

Detections for Page 17.png

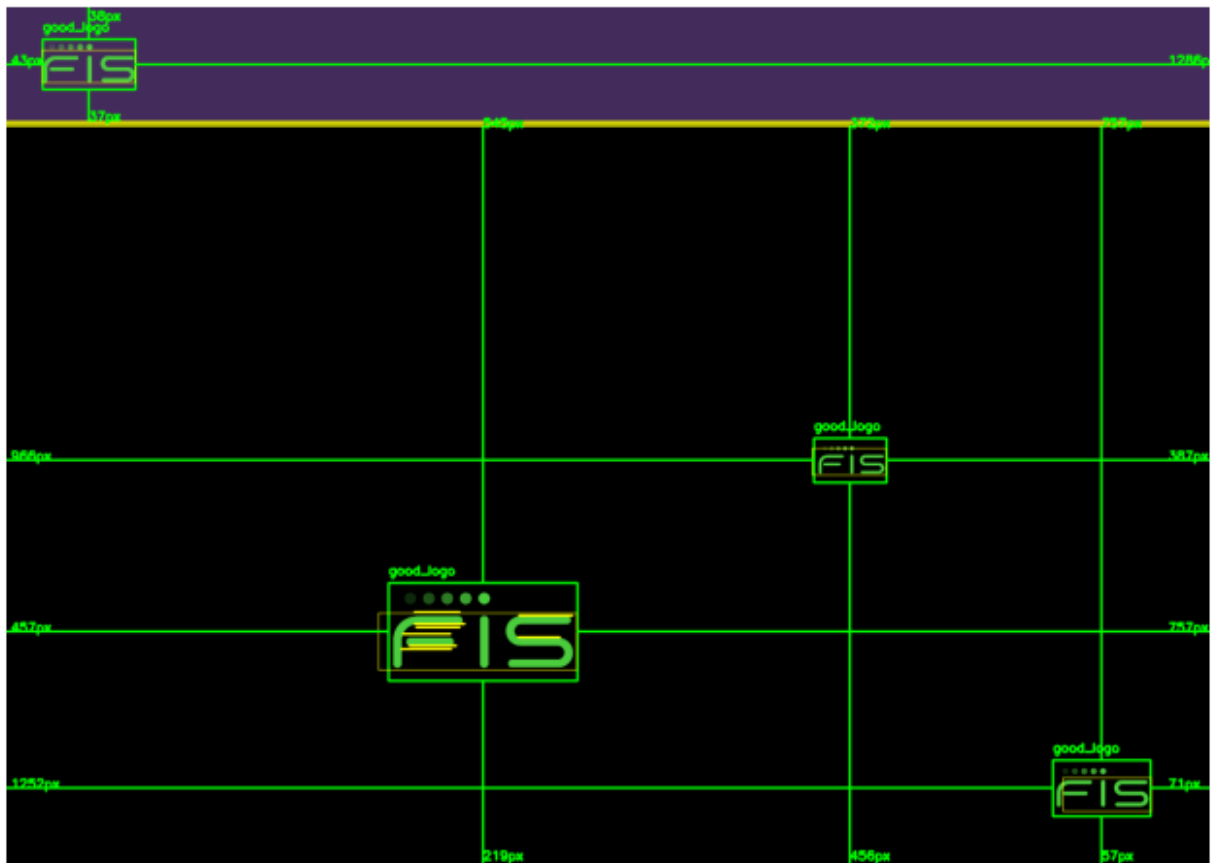


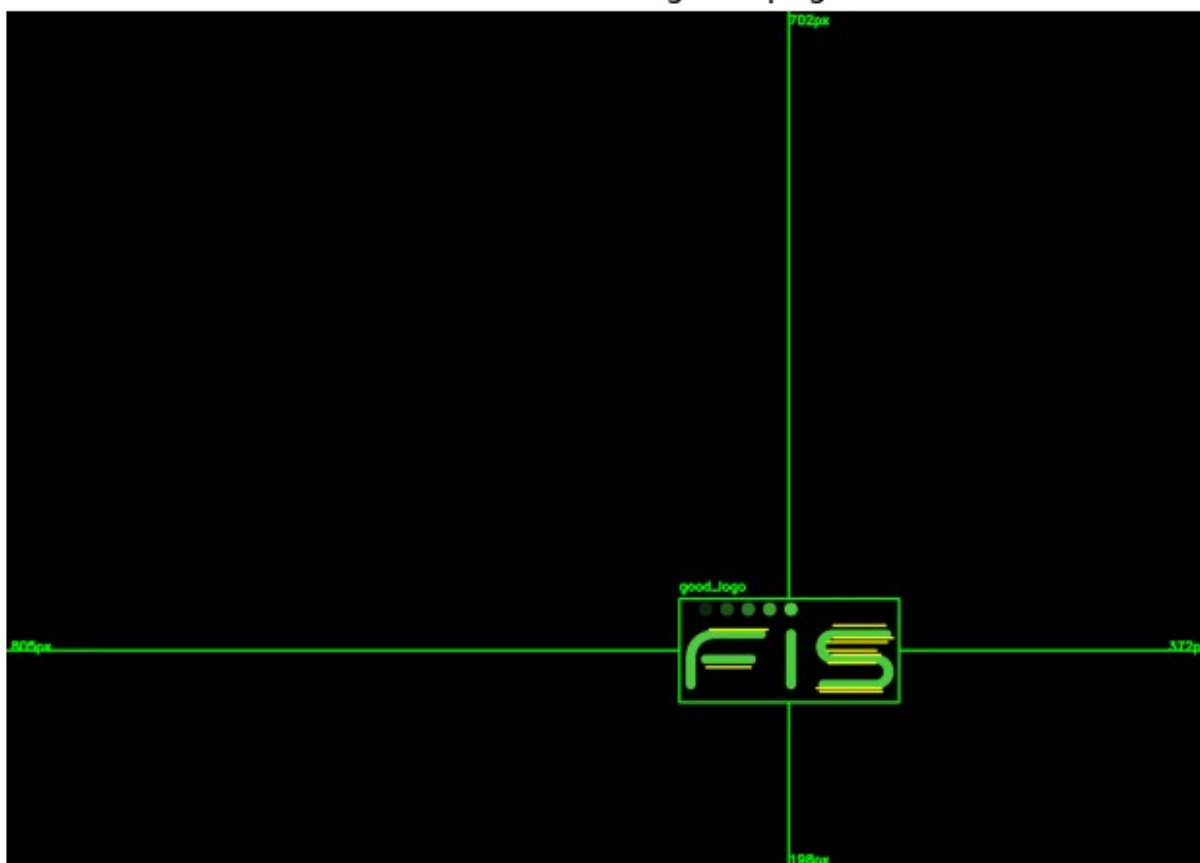
image 1/1 /content/test/Page 10.png: 480x640 1 bad_button, 2
bad_logos, 13.9ms
Speed: 4.9ms preprocess, 13.9ms inference, 1.6ms postprocess per image
at shape (1, 3, 480, 640)

Detections for Page 10.png



image 1/1 /content/test/Page 20.png: 480x640 1 good_logo, 13.0ms
Speed: 3.9ms preprocess, 13.0ms inference, 1.2ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 263 Minimum clearance: 75.14285714285714
Bounding Box (805, 702, 1068, 826) - Distances: {'Up': 702, 'Down':
198, 'Left': 805, 'Right': 372} - VALID

Detections for Page 20.png



```
image 1/1 /content/test/Page 6.png: 480x640 2 bad_buttons, 1 bad_logo,
1 good_disabled_button, 3 good_hover_buttons, 2 good_logos, 13.0ms
Speed: 2.7ms preprocess, 13.0ms inference, 1.5ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 112 Minimum clearance: 32.0
Bounding Box (69, 37, 181, 99) - Distances: {'Up': 37, 'Down':
np.int32(33), 'Left': 69, 'Right': 1259} - VALID
Bounding Box size: 80 Minimum clearance: 22.857142857142858
Bounding Box (1027, 803, 1107, 844) - Distances: {'Up': np.int32(664),
'Down': 180, 'Left': 758, 'Right': 333} - VALID
```

Detections for Page 6.png

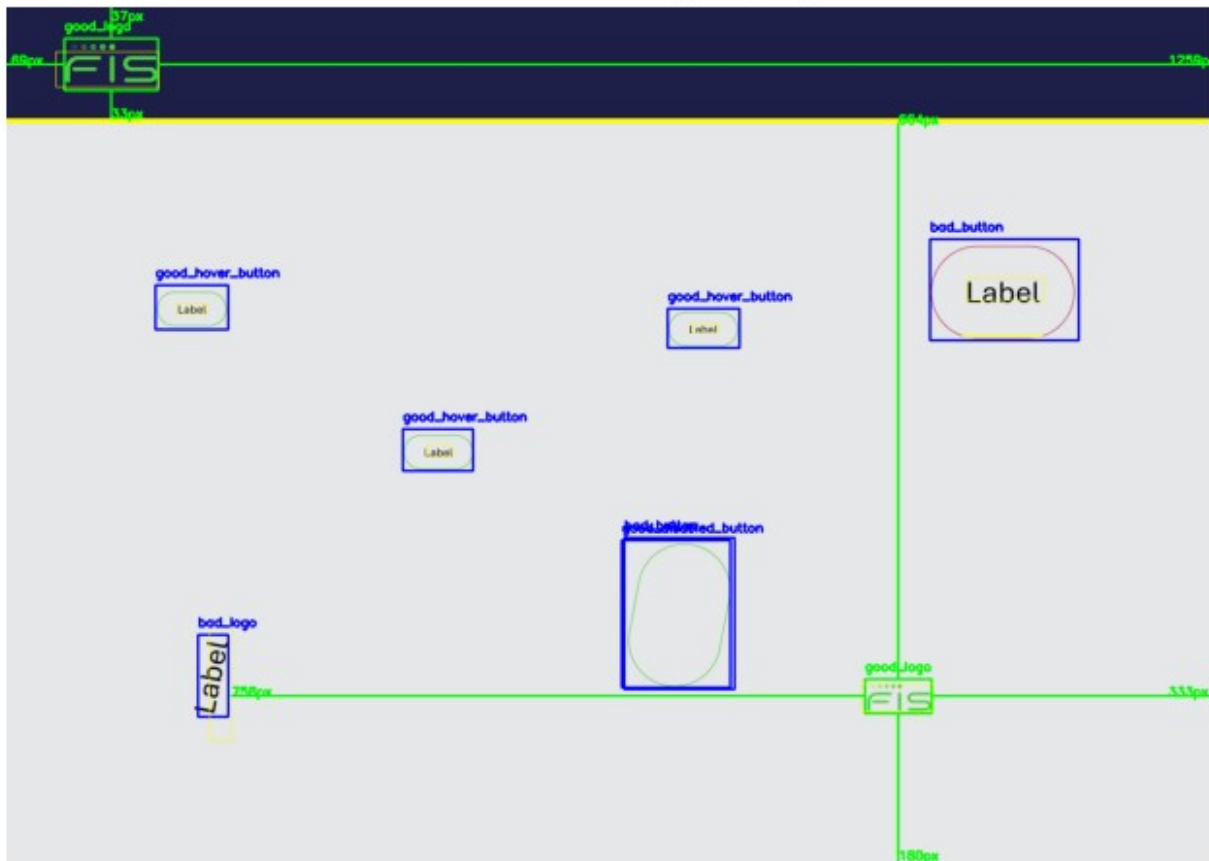


image 1/1 /content/test/Page 13.png: 480x640 1 bad_button, 2
good_logos, 13.0ms
Speed: 3.0ms preprocess, 13.0ms inference, 1.3ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 240 Minimum clearance: 68.57142857142857
Bounding Box (1087, 81, 1327, 197) - Distances: {'Up': 81, 'Down':
185, 'Left': 1087, 'Right': 113} - VALID
Bounding Box size: 110 Minimum clearance: 31.428571428571427
Bounding Box (44, 38, 154, 98) - Distances: {'Up': 38, 'Down':
np.int32(37), 'Left': 44, 'Right': 1286} - VALID

Detections for Page 13.png

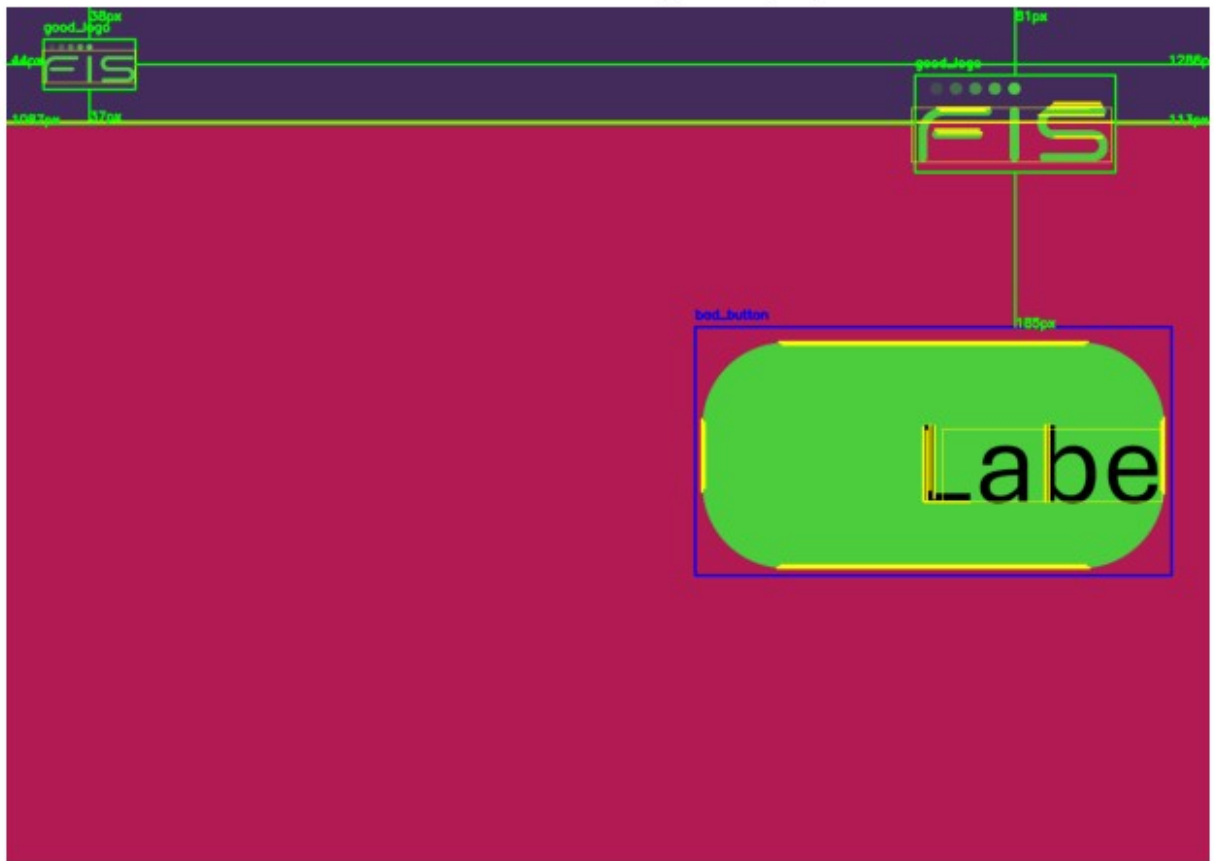


image 1/1 /content/test/Page 4.png: 480x640 4 bad_buttons, 1
good_disabled_button, 1 good_logo, 12.9ms
Speed: 2.9ms preprocess, 12.9ms inference, 1.4ms postprocess per image
at shape (1, 3, 480, 640)
Bounding Box size: 119 Minimum clearance: 34.0
Bounding Box (662, 242, 781, 297) - Distances: {'Up': 242, 'Down':
727, 'Left': 201, 'Right': 659} - VALID

Detections for Page 4.png

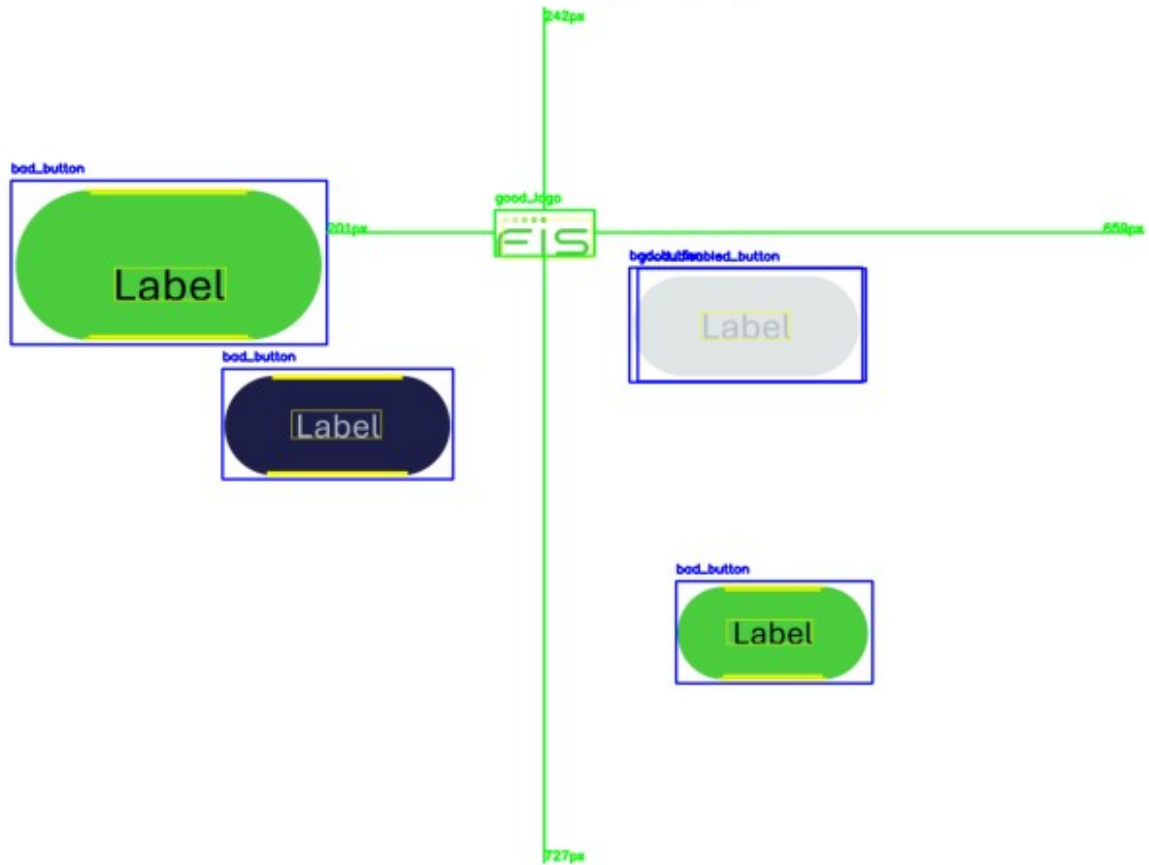


image 1/1 /content/test/Page 9.png: 480x640 1 bad_logo, 1 good_logo, 1 good_static_button, 13.0ms
 Speed: 4.6ms preprocess, 13.0ms inference, 1.8ms postprocess per image at shape (1, 3, 480, 640)
 Bounding Box size: 111 Minimum clearance: 31.714285714285715
 Bounding Box (43, 37, 154, 98) - Distances: {'Up': 37, 'Down': np.int32(37), 'Left': 43, 'Right': 1286} - VALID

Detections for Page 9.png



image 1/1 /content/test/Page 15.png: 480x640 1 bad_logo, 12.9ms
Speed: 2.8ms preprocess, 12.9ms inference, 1.3ms postprocess per image
at shape (1, 3, 480, 640)

Detections for Page 15.png

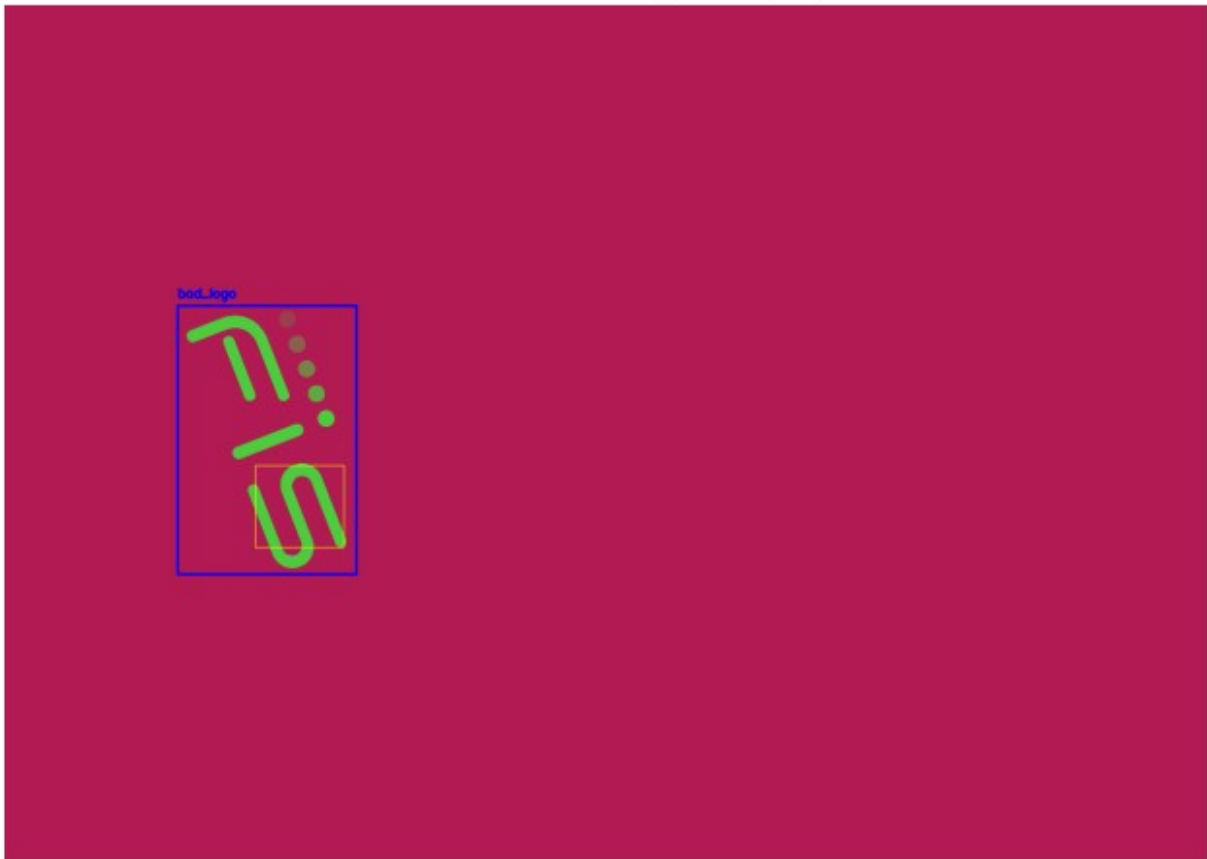


image 1/1 /content/test/Page 18.png: 480x640 1 bad_logo, 1 good_logo, 12.9ms
Speed: 2.7ms preprocess, 12.9ms inference, 1.3ms postprocess per image at shape (1, 3, 480, 640)
Bounding Box size: 112 Minimum clearance: 32.0
Bounding Box (199, 39, 311, 100) - Distances: {'Up': 39, 'Down': np.int32(35), 'Left': 199, 'Right': 1129} - VALID

Detections for Page 18.png

